

The Propagation of Environmental Risk Through Production Networks: Borrowing Cost Effects

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Abstract

We develop a general equilibrium model with firms, banks, and households to explain why firms more exposed to climate transition risk face higher borrowing costs. Firms are connected through production networks and make rigid input choices before shocks realize, generating endogenous default risk. Banks price debt competitively by accounting for expected defaults. As a result, borrowing costs depend not only on firms' direct exposure to transition risk but also on the one of their suppliers: even green firms may face higher financing costs when linked to brown upstream sectors. We calibrate the model using U.S. input-output linkages and observed borrowing-cost spreads between green and brown sectors, measuring exposure through carbon emissions. Importantly, we construct a model-consistent measure of network-adjusted carbon exposure by combining sectoral CO₂ emissions, input-output linkages, and U.S. firm-level financial data. We calibrate the model and perform regressions and diff-in-diff. Lenders price both direct and network-adjusted carbon exposure, consistent with the model's predictions. Evidence around the Paris Agreement supports a causal interpretation of transition-risk transmission through production networks.

JEL Codes: E44, G32, Q54, L14.

Keywords: Transition risk, Cost of debt, Production networks, Environmental shocks, Default risk, Carbon emissions.

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1 Introduction

Firms are subject to climate physical risks, or losses due to natural disasters, and transition risks, due to regulation, reputation damages and technological obsolescence. Brown firms, usually identified with more polluting ones, are, by their nature, more exposed to transition risks. The question of how physical and transition risks, that we name environmental or climate risks, are transmitted into financial markets is central to both policy debates and academic research.

There is growing evidence that firms with larger carbon footprints or similar measures of brownness face higher loan costs, both in the US and outside (see for instance [Ivanov et al. \(2023\)](#), [Kacperczyk and Peydro' \(2024\)](#), [Delis et al. \(2024\)](#), [Altavilla et al. \(2023\)](#), [Ehlers et al. \(2022\)](#), [Reghezza et al. \(2022\)](#)). A similar evidence holds for bonds (see [Zerbib \(2019\)](#) and [Baker et al. \(2018\)](#), among others).

These findings have been taken as an indication that credit markets begin to internalize physical as well as transition risks. Yet the existing evidence, with the exception of [Croce et al. \(2026\)](#) and [Kou et al. \(2026\)](#), remains reduced-form and partial: it interprets climate risk as a firm-level attribute, mostly proxied by direct emissions, and studies its correlation with observed spreads. Such an approach neglects two essential aspects. First, firms do not operate in isolation, but are embedded in complex input-output networks, where shocks originating in one sector propagate to others. Second, interest rates are not exogenously given risk premia, but equilibrium outcomes that result from the interaction of production, financing, and default risk. A proper understanding of environmental risks propagation and financial market pricing requires a framework that makes these mechanisms explicit.

This paper is an attempt to develop such a framework. Building on recent advances in the theory of production networks, we construct a static general equilibrium model with three types of agents: firms, consumers, and banks. Firms are heterogeneous in their environmental exposure. Green firms are subject only to physical risks, while brown firms are also exposed to transition risks. In the absence of shocks, brown firms may be more productive than green ones. Crucially, firms must decide ex ante on inputs and labor without being able to adjust to the realized state. This rigidity, reflecting real-world delays and irreversibilities in production and employment, generates the possibility of default when shocks realize, or ex post.

Shocks do not stop at the firm level. Because production uses intermediate goods, shocks propagate through input-output linkages. The Leontief inverse provides the natural tool

to characterize this propagation: the primitive shocks to productivity are transformed into network-adjusted shocks that affect all downstream firms. Hence, a firm’s effective exposure is determined not only by its own environmental risk but also by that of its suppliers.

Banks operate under perfect competition, are risk-neutral, and finance firms’ liabilities. They are residual claimants, paid only after workers and suppliers, and are therefore directly exposed to default risk. Interest rates are set so that banks break even in expectation. This zero-profit condition delivers a simple but powerful characterization: the interest rate applied to any firm depends on the distribution of its total shocks, namely the primitive ones affecting its own output and the ones inherited from its suppliers. Productivity differences between green and brown firms do not directly affect borrowing costs; only risk does.

The model yields several implications. First, firm borrowing costs increase not only with direct exposure but also with the exposures of upstream suppliers. Thus even firms with low direct emissions may face high interest rates if they are embedded in brown supply chains, because primitive shocks to brown firms are greater than those to green firms. Second, interest rates on brown firms are weakly higher if the network is the same for brown and green firms, or if brown firms are more connected to brown and green to green. These results show how risk, network propagation, and endogenous pricing jointly determine the allocation of credit.

We bring the mechanisms highlighted by the model to the data by assembling a novel panel that combines U.S. firm-level balance sheet information and borrowing costs with sector-level carbon emissions from the Environmental Protection Agency and detailed input-output linkages from national accounts. This empirical design allows us to construct both a measure of direct carbon exposure and a model-consistent measure of network-adjusted emissions that captures firms’ inherited exposure to transition risk through production networks.

We first calibrate the model to data and show that it matches the spread between the borrowing costs of green and brown sectors in the US manufacturing industry. The ability of the model to match the spread relies on both the original exposure to shocks and the greater interconnectedness among green sectors on the one side, brown on the other, without perfect separation.

Our subsequent regression analysis delivers two main findings. First, we show that exposure to carbon emissions is positively associated with firms’ borrowing costs. When considered separately, both direct CO₂ emissions and network-adjusted CO₂ emissions are associated with significantly higher interest rates. This indicates that credit markets price

climate-related risk and that firms operating in more emission-intensive environments face systematically higher financing costs.

Second, and more importantly, we show that both direct emissions and network-adjusted emissions matter once we isolate the residual component of the network measure. When direct CO₂ emissions are included together with the residualized network exposure, both coefficients remain economically large and statistically significant. This indicates that the network measure captures variation beyond firms’ own emission levels.

We further validate this mechanism by exploiting the 2016 entry into force of the Paris Agreement as an exogenous shift in climate policy expectations. Using difference-in-differences and event-study designs, we show that borrowing costs increase disproportionately for firms embedded in carbon-intensive production networks following the Agreement. The absence of differential pre-trends and the concentration of post-treatment effects in the network-based measure provide causal support for the interpretation that lenders revise credit pricing in response to anticipated transition risks transmitted through supply chains.

Taken together, the theory and evidence show that climate transition risk is priced in credit markets primarily through production networks. Borrowing costs reflect not only firms’ own environmental footprint, but, more fundamentally, their exposure to upstream carbon risk and their position in the supply chain. The empirical results therefore strongly support the model’s central prediction that transition risk operates as a systemic, network-amplified phenomenon rather than a purely firm-level attribute.

The remainder of the paper is structured as follows. Section 2 develops the theoretical model, introducing rigid production, environmental shocks, and competitive banks, and derives the main results on how climate-transition risk and production-network linkages shape equilibrium borrowing costs. Section 3 describes the data and explains the construction of our measures of direct and network-adjusted carbon exposure, combining firm-level financial information, sectoral emissions data, and input–output linkages. Section 4 calibrates the model using U.S. data and evaluates its ability to reproduce the observed borrowing-cost differential between green and brown sectors. Section 5 presents the empirical analysis, testing the model’s predictions and exploiting the Paris Agreement as a quasi-natural experiment to identify the transmission of transition risk through production networks. Section 6 concludes.

Literature. The motivation of our paper is in the largely documented evidence on the different credit conditions applied to green and brown firms.

Empirically, a growing body of literature indeed shows that environmental performance affects corporate financing. In credit markets, [Ivanov et al. \(2023\)](#) show that in the US borrowing conditions from banks, including interest rates, are more severe for brown firms. This happens after both the California cap-and-trade act and the Waxman-Markey act. [Kacperczyk and Peydro’ \(2024\)](#) use a sample of global firms to show that, once banks commit to decarbonization, polluting firms receive less credit. A similar evidence has been detected in the EU banking loans. [Altavilla et al. \(2023\)](#) demonstrate that euro-area banks charge higher interest rates to firms with greater carbon emissions and lower rates to firms committing to emissions reductions, even after controlling for default risk. These effects are more pronounced among banks publicly committed to decarbonization. Restrictive monetary policy amplifies both credit risk premia and emission-related pricing premia, tightening credit more for brown firms than for green ones. Related evidence from [Ehlers et al. \(2022\)](#) shows that polluting firms began paying a “carbon premium” on syndicated loans after the Paris Agreement. Also [Delis et al. \(2024\)](#) find that after the Paris Agreement, banks began pricing climate policy risk into syndicated loans, especially for firms with large fossil fuel reserves.

More broadly, earlier studies had already established that firms with poor CSR or environmental records face higher borrowing costs. [Goss and Roberts \(2011\)](#) show that socially irresponsible firms pay a premium on bank loans, while [Chava \(2014\)](#) document higher loan spreads for environmentally controversial firms. In bond markets, a modest “greenium” has been identified: [Zerbib \(2019\)](#) and [Baker et al. \(2018\)](#) find that green bonds carry slightly lower yields relative to comparable conventional bonds, consistent with investor preference channels highlighted by [Pastor et al. \(2021\)](#).

As a result, [Gormsen and Huber \(2023\)](#) show that since 2016 greener firms have enjoyed a structurally lower capital cost, roughly 1% lower than brown firms, reflecting both the equity and debt markets.

Relative to the existing literature, our contribution is to develop a general equilibrium model with financial frictions and endogenous default in which transition shocks affect equilibrium borrowing costs. We show that the cost of debt depends not only on firms’ direct exposure to transition risk, but also on the amplification of shocks through production networks. In our framework, upstream disruptions generated by transition shocks propagate through supply chains, affecting downstream firms’ revenues, default probabilities, and financing conditions. As a result, even firms with low direct emissions may face higher borrowing costs because of their exposure to carbon-intensive suppliers.

To formalize the role of shock transmissions thorough input-output chains, we draw on the

literature on production networks, along the seminal contribution of [Acemoglu et al. \(2012\)](#). We build on the frameworks of [Como et al. \(2026\)](#) and [Pellet and Tahbaz-Salehi \(2023\)](#), who introduce rigidity in input contracting. Rigidity means that orders for intermediate goods and investments occur before shocks to production realize, and cannot be adjusted when the true state of the world becomes known, as we witnessed during the Covid-19 pandemic and recent disruptions from wars. Because of rigidities, firms suffer losses and may even default. Relative to those papers, here we distinguish green and brown firms based on the risks they suffer and the productivity they enjoy. In [Pellet and Tahbaz-Salehi \(2023\)](#) there was neither leverage nor green vs brown firms, in [Como et al. \(2026\)](#) there was leverage but no distinction between green and brown firms.

[Croce et al. \(2026\)](#) develop a network-based methodology to measure embodied emissions and Scope-3 exposure along global supply chains using input-output linkages and firm-level network data. Their framework highlights how carbon exposure propagates internationally through production networks and provides a general-equilibrium (GE) approach to measuring emissions embedded in upstream relationships. They also include debt, but build on a GE models without rigidities, their main objective being the construction of the aggregated network CO₂ measure, which we developed independently.

Relatedly, [Kou et al. \(2026\)](#) develop a GE asset-pricing model in which carbon risk propagates through input-output networks. They show that indirect network exposure is a key determinant of the carbon risk premium and that firms with low direct emissions may nevertheless bear substantial transition risk because of their supply-chain connections. Their paper therefore complements our, since we look at debt while they look at equity pricing.

2 Model

The theoretical model builds on [Como et al. \(2026\)](#). We develop a static general equilibrium in a production network with three types of agents: firms, consumers, and banks. All decisions are taken at time 0, prior to the realization of uncertainty. At time 1, the stochastic shocks on production are realized, production occurs, consumers consume, and payments between agents are settled. Uncertainty in production is introduced by defining a complete probability space $(\Omega, \mathcal{F}, P)$, on which random shocks are realized. Agents make ex-ante decisions at time 0, while outcomes and transfers materialize ex-post, at time 1.

2.1 Rigidity

Production processes in the real world are characterized by lags and irreversibilities. Orders of intermediate goods and investment in physical capital typically take time to be delivered, and therefore, even when there is uncertainty on the future state of the economy, are planned in advance, without the possibility of being tailored to the state of the world that will realize. When a specific state occurs, these decisions are often irreversible. Similarly, labour is often hired in advance and cannot promptly be dismissed, at least in some jurisdictions. These forms of *rigidity*, which have recently been introduced in the network literature (Pellet and Tahbaz-Salehi (2023)), may generate default, if firms are levered. If, in a specific state, the costs of orders for intermediate goods and labour hired, as well as the interest rates to banks, are not covered by revenues, default occurs. To model all of this, we assume that the decision of firms about intermediate goods and labour are not state dependent. They are characterized by ex ante or *nominal* quantities, constant across possible states.

2.2 Firms

There are n firms or sectors in the economy⁰, $\mathcal{V} = 1, 2, \dots, n$, partitioned into two groups. The first n_1 firms are classified as *green*, while the remaining $n - n_1$ firms are classified as *brown*. Brown firms are more risky than green, in that they can be hit by transition as well as physical shocks, but are more productive, all others equal. Transition shocks include in our view fines, penalties, and other reputational and regulatory risks, as well as technological obsolescence. Physical risks correspond to natural catastrophes and their consequences.

On the probability space $(\Omega, \mathcal{F}, P)$, we define two vectors of non-positive random variables: $\lambda : \Omega \rightarrow \mathbb{R}_-^n$ and $\epsilon : \Omega \rightarrow \mathbb{R}_-^n$, whose first n_1 components are equal to zero. These represent the primitive production log shocks, for brevity the *primitive production shocks*. Green firms are exposed only to λ , which captures physical risks. Brown firms are exposed to both λ and an additional transition shock ϵ . Let $\eta = \lambda + \epsilon$ denote the overall vector of primitive shocks. The previous assumptions on the components of η are equivalent to stating that shocks to brown firms are larger in absolute value than those to green firms, or that *green shocks dominate brown ones in the sense of first-order stochastic dominance*. Let G_k

⁰We use indifferently the term firm and sector because we assume that firms are perfect substitutes: each sector is made by homogeneous firms, and as one firm defaults it can be substituted by a perfectly equal one. This prepares the empirical part, since the US input-output matrix is available to us at the 6 digit sector and not at the individual firm level

be the cumulative distribution function of $|\eta_k|$: for any $b > n_1$ (brown) and $g \leq n_1$ (green),

$$G_b(x) \leq G_g(x) \quad \text{for all } x \in \mathbb{R}^n, \text{ with strict inequality for some } x, \quad (1)$$

When considering η_k instead of its absolute value, since shocks are non-positive, the inequality reverses. If F_k denotes the cumulative distribution function of η_k :

$$F_b(x) \geq F_g(x), \quad \text{for all } x \in \mathbb{R}^n, \text{ with strict inequality for some } x, \quad (2)$$

The definition of transition and physical risks translates into the following assumption: $\eta_g \geq \eta_b$ in the sense of first-order stochastic dominance. Considering the absolute value of the shocks, $-\eta_g \leq -\eta_b$: later on, we label this saying that *green shocks are smaller than brown*.

To capture the basic trade-off between productivity and risk, and anticipating on the empirical evidence, we assume that brown firms, in the absence of shocks, are more productive than green ones. Let $Z \in \mathbb{R}_+^n$ denote the vector of total factor productivities. We assume Z_k is higher for brown firms ($k > n_1$) and lower for green firms ($k \leq n_1$), reflecting a higher baseline output among brown firms that compensates for their greater environmental exposure.

Firms use a Cobb-Douglas technology to produce output:

$$y_k^\eta = Z_k e^{\eta_k} l_k^{\beta_k} \prod_{j \in \mathcal{V}} (z_{jk}^\eta)^{A_{jk}}, \quad \text{for } k = 1, \dots, n, \quad (3)$$

where:

- y_k^η is the output of firm k after the shock η ,
- $Z_k > 0$ is the firm-specific productivity,
- l_k is labor employed,
- $\beta_k \geq 0$ is the labor share,
- z_{jk}^η is the input of good j used by firm k after the shock η ,
- $A_{jk} \geq 0$ reflects the *importance* of input j in the production of firm k .

The case of no shocks is denoted by $\eta = 0$. We define $\alpha_k = \sum_{j \in \mathcal{V}} A_{jk}$ and assume constant returns to scale:

$$\alpha_k + \beta_k = 1. \quad (4)$$

Let c_k denote the consumption of good k .

We collect output, consumption, labor, and productivity in the vectors $y, c, l \in \mathbb{R}_+^n, Z \in \mathbb{R}_{++}^n$, and the input coefficients z_{jk} in a matrix $z \in \mathbb{R}_+^{n \times n}$. The production function of firm k can be compactly written as $y_k = F_k^{\eta_k}((z_{jk})_j, l)$, where $(z_{jk})_j$ is column k of the matrix z , which collects all the inputs to sector k .

The market clearing conditions are:

1. Goods market clearing:

$$y = z\mathbf{1} + c, \quad (5)$$

2. Labor market clearing:

$$l'\mathbf{1} = 1. \quad (6)$$

where $\mathbf{1}$ is a vector of ones.

We also collect the coefficients A_{jk} into a matrix $A \in \mathbb{R}_+^{n \times n}$. Under the mild assumption that the spectral radius of $A, \sigma(A)$ be smaller than one, the matrix A defines a Leontief inverse:

$$L = (I - A')^{-1}, \quad (7)$$

which has non-negative entries and plays a central role in shock propagation throughout the network, as we show now.

We assume that shocks propagate proportionally through the network: if a firm is hit by a shock that reduces output by a fraction p , it also reduces its supply of intermediate goods equally across suppliers and its supply of final goods by the same fraction. Formally, for all k, j such that $A_{kj} > 0$,

$$\frac{\tilde{y}_k}{y_k} = \frac{\tilde{z}_{kj}}{z_{kj}} = \frac{\tilde{c}_k}{c_k}. \quad (8)$$

where the values with a tilde are the actual or post-shock ones, while the ones without the tilde are the ex ante or nominal ones. Como et al. (2025) show that, under this assumption, the actual shocks received by firms are not the primitive shocks η , but the network-amplified shocks $\rho = L\eta$. A shock η_j originating in node j affects firm k if $L_{kj} > 0$:

$$\rho_k = \sum_{j \in \mathcal{V}} L_{kj} \eta_j. \quad (9)$$

The vector ρ has non-positive entries, which represent the total shocks to the different sectors, opposite to the primitive ones in η . This highlights the core network effect: produc-

tion shocks can be magnified or attenuated depending on the structure of interfirm linkages. A sufficient condition for the total shocks to brown firms to be greater than the ones of green ones - as it happened for primitive ones - is that the Leontief matrix separates the two networks, or $L_{kh} > 0$ when both firms are either green or brown ($k \leq n_1, h \leq n_1$ and $k > n_1, h > n_1$), while $L_{kh} = 0$ in the mixed cases, when one is green and the other brown.

Lemma 1. *If $L_{kh} > 0$ for $k \leq n_1, h \leq n_1$ and $k > n_1, h > n_1$, $L_{kh} = 0$ otherwise, $\rho_g \geq \rho_b$ in the sense of first-order stochastic dominance.*

To complete the description of the firm side of the economy, we introduce

- w , unit cost of the employed labor (*wage*);
- p_k , unit price of the good produced by firm k .

Because of (8), the *actual* assets or revenues of firm k are

$$\mathcal{A}_k := p_k y_k^\eta = p_k e^{\rho_k} F_k^0((z_{jk})_j, l_k) \quad (10)$$

On the other hand, its *actual* liabilities due to intermediate goods and labor are

$$\mathcal{L}_k := \sum_{j \in \mathcal{V}} p_j z_{jk}^\eta + w l_k = \sum_{j \in \mathcal{V}} p_j e^{\rho_j} z_{jk} + w l_k. \quad (11)$$

Firms raise debt from financiers, which we call banks. As we explain below, the bank finances a fraction θ_k of the liabilities., so that θ_k represents the *leverage ratio* of firm k . The bank either receives the full payment of the loan and its interests, $(1 + r_k)\theta_k \mathcal{L}_k$, or gets only a recovery out of it. In the second case the firm loses the complement of the recovery with respect to the full payment, $(1 + r_k)\theta_k \mathcal{L}_k$, as default costs. So the firm always pays $(1 + r_k\theta_k)\mathcal{L}_k$, either to workers and to suppliers or as default costs. Then, firms' profits are:

$$\Pi_k((z_{jk})_j, l_k, \eta, p) := \mathcal{A}_k - (1 + r_k\theta_k)\mathcal{L}_k \quad (12)$$

They can take any sign: for the sake of simplicity, we name them profits also when they are negative.

Actual assets, liabilities, and profits can be expressed in terms of normalized quantities, as follows. Define the *normalized total shocks* as the total shocks which affect k , divided by

their expectation under the measure P

$$\tau_k := \frac{e^{\rho_k}}{\mathbb{E}[e^{\rho_k}]}. \quad (13)$$

Define also the *normalized suppliers' shocks* as the normalized total shocks to the nodes who provide inputs to node k , weighted by their importance. Considering that log shocks to labor are zero by assumption and inputs to sector k include labor, with importance β_k , the suppliers' shocks are

$$\epsilon_k := \beta_k + \sum_{j \in \mathcal{V}} A_{jk} \tau_j. \quad (14)$$

Then, defining $s_k := \mathbb{E}[\mathcal{A}_k]$

$$\mathcal{A}_k = s_k \tau_k \quad (15)$$

$$\mathcal{L}_k = s_k \frac{\epsilon_k}{1 + r_k \theta_k}. \quad (16)$$

$$\Pi_k = s_k (\tau_k - \epsilon_k). \quad (17)$$

2.3 Banks

Financiers form a continuum of risk-neutral banks operating under perfect competition. Each bank is randomly matched with a firm and provides financing to cover a fraction $\theta_k \in [0, 1]$ of the firm's liabilities. Banks are paid only after firms have settled payments to workers and suppliers. Because of that, they are exposed to the possibility of default, which arises endogenously from the realization of shocks.

The bank's profit from lending to firm k is:¹

$$\mathcal{I}_k = \min(\max(\mathcal{A}_k - (1 - \theta)\mathcal{L}, 0), (1 + r)\theta_k \mathcal{L}_k) - \theta_k \mathcal{L}_k \quad (18)$$

$$= \min(\max(\mathcal{A}_k - \mathcal{L}, -\theta_k \mathcal{L}_k), r_k \theta_k \mathcal{L}_k). \quad (19)$$

Because banks operate competitively and are risk-neutral, they choose the interest rate r_k so as to break even in expectation. The zero-profit condition is:

$$\mathbb{E}[\mathcal{I}_k] = 0 \quad (20)$$

¹The payoff to bank k is the one described here if $\mathcal{A}_k \geq 0$, namely if its revenues are non-negative, i.e. the financed firm does not sell negative quantities at positive prices or positive quantities at negative prices. Since the only equilibrium features positive prices and quantities of goods, for every θ , the requirement is satisfied as a strict equality: $\mathcal{A}_k > 0$. The reader interested in the general case ($\mathcal{A}_k \geq 0$ as well as $\mathcal{A}_k < 0$), and in the proof that also this more general case leads to positive revenues, can look at [Como et al. \(2026\)](#).

To simplify matters, in what follows we consider only the case in which liabilities are totally financed by debt, or $\theta = 1$. In that case, the condition that determines the interest rate for sector k becomes

$$\mathbb{E}[\min(\mathcal{A}_k - \mathcal{L}_k, r_k \mathcal{L}_k)] = \frac{s_k}{1 + r_k} \mathbb{E}[\min(\tau_k(1 + r_k) - \epsilon_k, r_k \epsilon_k)]. \quad (21)$$

$$\mathbb{E}\left[\frac{-\epsilon}{1 + r} + \min(\tau_k, \epsilon_k)\right] = 0. \quad (22)$$

and finally, since $\mathbb{E}[\epsilon_k] = 1$

$$\mathbb{E}[\min(\tau_k, \epsilon_k)] = \frac{1}{1 + r_k}. \quad (23)$$

Therefore, the interest rate applied to each firm depends exclusively on its exposure to environmental shocks, and not directly on its productivity² Z_k , even though brown firms are assumed to be more productive than green firms (i.e., $Z_i > Z_j$ for $i > n_1$ and $j \leq n_1$). Interest rates depend only on shocks, because default does. However, and consistently with the transmission of shocks from one firm to the other through the network, the shocks that matter are not only the primitive ones, η , but the total ones, and specifically the total shocks to the sector and to its suppliers, τ and ϵ .

Interest rates are non negative, since $\mathbb{E}[\min(\tau_k, \epsilon_k)] \leq \mathbb{E}[\epsilon_k] = 1$.

2.4 Network effects on the cost of debt

In this section we study the relationship between risks and cost of debt. We show that, in equilibrium, the cost of debt on green firms is smaller than the one on brown ones in a number of cases. First, when the network linkages are uniform, in that all sectors have the same importance to the other ones. Second, if the networks formed by green and brown firms are almost separated (brown firms are more important to brown, green to green) and third, if the two subnetworks are completely separated.

Let us consider first the case in which labour - which is riskless - is more important for green than for brown firms and all the importance coefficients - contained in the matrix A - are the same, $A_{jg} = A_{jb}$ for all indices j, g, b . This is an economically interesting case not only because all intermediate goods are equally important, be them produced or purchased by green or brown firms. But also because the transmission of the shocks in the network occurs

²The results of the theoretical model are therefore robust with respect to the higher productivity assumption of brown firms. Interest rates of brown firms, under the conditions of the theorems below, would be greater than the green ones also in the absence of such an assumption.

in a uniform way. A matrix A of this type indeed implies that the off-diagonal terms of the Leontief matrix are the same: transmission of the primitive shocks to any other sector occurs in the same, *uniform* way. It is not difficult to guess that, in such a “neutral” transmission environment, brown sectors, whose primitive shocks are more severe, but are affected in the same way as green sectors by the shocks of other sectors, remain more risky and therefore deserve a greater interest rate. Banks charge higher interest rates to brown firms in order to compensate for higher default risk. This is the content of the next Theorem, which is proven in Appendix A.1.

Theorem 1. *[Interest Rates with a uniform network] If green firms do not employ less labour than brown firms, i.e., $\beta_g \geq \beta_b$ for all $g \leq n_1$ and $b > n_1$, and $A_{jg} = A_{jb}$ for all indices j, g, b , then the interest rate applied to brown firms is weakly higher than that applied to green firms:*

$$r_b \geq r_g, \quad (24)$$

.

The theorem outcome results from the brown firms’ higher exposure to both physical and transition risks, as a result of their direct - not network related - exposure.

The second circumstance we study is the one in which green firms not only are affected by less severe primitive shocks, but are also more exposed to shocks transmitted by other green firms, because the importance of green firms for them is higher than the one of brown firms. The intuition is that the networks are *almost separated*. The following theorem is also proved in Appendix A.1:

Theorem 2. *[Interest Rates with stronger linkages among green vs brown sectors] If green sectors are more important than brown ones in the production of green sectors, $\sum_{j \leq n_1} A_{gj} \geq \sum_{j \leq n_1} A_{bj}$ for all indices $g \leq n_1, b > n_1$, then the interest rate applied to brown firms is weakly higher than that applied to green firms:*

$$r_b \geq r_g, \quad (25)$$

The extreme case of the previous network separation is the one in which the networks are *completely separated*, which is described by the hypotheses of Lemma 1.³ Based on the

³This is the circumstance which would be most welcome, at least in Europe, by the regulatory authority, in order to have very favourable loan conditions for green firms (see for instance [European Union \(2024\)](#)). We briefly discuss this in the conclusions

previous Theorem, it would give the minimum interest rate for green firms. The following Corollary is also proved in Appendix A.1:

Corollary 1. *[Interest Rates with separated networks] If green firms do not employ less labour than brown firms, i.e., $\beta_g \geq \beta_b$ for all $g \leq n_1$ and $b > n_1$, and the hypotheses of lemma 1 hold, then the interest rate applied to brown firms is weakly higher than that applied to green firms:*

$$r_b \geq r_g, \quad (26)$$

As a further corollary, it is easy to prove that, if there are no physical risks, green firms deserve a zero interest rate, because they run no default risk at all. Also in this case the proof is in Appendix A.1.

Corollary 2. *[Interest Rates with separated networks and no physical risks] If green firms do not employ less labour than brown firms, i.e., $\beta_g \geq \beta_b$ for all $g \leq n_1$ and $b > n_1$, the hypotheses of lemma 1 are satisfied and there are no physical risks ($\lambda = 0$), then the interest rate on green sectors is zero.*

All these results prove that risk-adjusted interest rates arise endogenously and reflect primitive environmental uncertainty, but also the way in which it is transmitted through the network.

2.5 Consumers

Consumers form a continuum and are modeled as a representative price-taking agent endowed with preferences over the n consumption goods. Preferences are described by a Cobb-Douglas utility function:

$$U(c) = \prod_{k \in \mathcal{V}} c_k^{\gamma_k}, \quad (27)$$

where $\gamma_k \geq 0$ is the consumer's weight on good k , and, in vector notation, $\gamma' \mathbf{1} = 1$. Because the coefficients of the utility function can also be smaller for green sectors than for brown ones, or can be all equal, preferences do not take into account any specific preference for greenness.

Like firms, the representative consumer makes a *rigid* nominal consumption plan at time 0, before the realization of shocks. Let $c = (c_k)_{k=1}^n$ denote this vector of nominal consumption choices. Once the shocks η are realized, because of (8), the actual consumption is scaled by the total shock affecting each good:

$$c_k^\eta = e^{\rho_k} c_k. \quad (28)$$

The consumer maximizes expected utility over the distribution of shocks:

$$\max_c \mathbb{E}[U(c^\eta)] \quad (29)$$

subject to a state-dependent budget constraint. The consumer receives:

- wages w from labor,
- profits or losses Π_k from firms, which include default costs,
- and interest income $r_k \theta_k \mathcal{L}_k$ from lending via banks, since consumers own both firms and banks.

The total endowment is therefore:

$$E = w + \sum_{k \in \mathcal{V}} \Pi_k + \sum_{k \in \mathcal{V}} r_k \theta_k \mathcal{L}_k. \quad (30)$$

The consumer's budget constraint, accounting for the realized shock to consumption, is:

$$\sum_{k \in \mathcal{V}} e^{\rho_k} c_k p_k \leq E. \quad (31)$$

Finally, define the (*Bonacich*) *network centrality vector* as:

$$\nu = L' \gamma. \quad (32)$$

Each element v_k of this vector weights consumer preferences through input-output linkages. It therefore measures the centrality of the corresponding sector in the consumer utility, taking into consideration the use of its goods for intermediate production of other goods, and the role of the latter in consumer's welfare.

2.6 Theoretical Framework: Equilibrium

We now characterize the general equilibrium (GE) of the economy described above. Following Como et al. (2026), we define a rigid Walrasian equilibrium as follows:

Definition 1 (Rigid Walrasian Equilibrium). Consider a Cobb-Douglas economy $(\mathcal{V}, A, \beta, \gamma, Z)$ satisfying (4), equipped with an interest rate vector r and a leverage vector θ . Given a distribution P over primitive log-productivity shocks η , a *rigid Walrasian equilibrium* is a tuple (y, z, c, l, p, w) such that:

- (i) For every firm $k \in \mathcal{V}$, the labor l_k and input bundle $(z_{jk})_j$ maximize expected profits $\mathbb{E}[\Pi_k]$ given prices p , wage w , interest rates r , and leverage θ .
- (ii) The consumption bundle c maximizes expected utility $\mathbb{E}[U(c^\eta)]$, subject to the state-dependent budget constraint:

$$\sum_{k \in \mathcal{V}} e^{\rho_k} c_k p_k \leq E. \quad (33)$$

We also need some additional notation. First, considering that $\theta_k = 1$, define the cost of debt under continuous compounding, instead of discrete, as

$$\zeta_k = \log(1 + r_k) \quad (34)$$

This reflects the cost of debt to the single firm k . Define the total cost of debt over the supply chain, as the sum of the costs over the chain, weighted by the corresponding Leontief coefficients:

$$\xi_k = \sum_{j \in \mathcal{V}} L_{kj} \zeta_j. \quad (35)$$

Also, let us define the Leontief inverse obtained from the importance coefficients modified because of leverage,

$$\bar{A}_{jk} := A_{jk} \frac{1}{1 + r_k \theta_k}, \quad \bar{L} := (I - \bar{A}')^{-1}, \quad (36)$$

Modify accordingly also the labour coefficients, and use them to redefine the Bonacich centrality with leverage

$$\begin{aligned} \bar{\beta}_k &:= \beta_k \frac{1}{1 + r_k \theta_k} \\ \nu^\zeta &:= \frac{\bar{L}' \gamma}{\bar{\beta}' \bar{L}' \gamma}, \end{aligned} \quad (37)$$

The new Bonacich vector conveys the same message of the one without leverage, it measures centrality vs consumers. The following result is adapted from Como et al. (2026):

Theorem 3 (Existence and Uniqueness). *Consider a Cobb-Douglas economy $(\mathcal{V}, A, \beta, \gamma, Z)$ satisfying (4). Then a unique rigid Walrasian equilibrium (y, z, c, l, p, w) exists, and satisfies:*

- **nominal productions:**

$$y_k^0 = \nu_k^\zeta e^{-\xi_k}; \quad (38)$$

- **nominal intermediate quantities:**

$$z_{jk}^0 = \nu_k^\zeta A_{jk} e^{-\zeta_k - \xi_j}; \quad (39)$$

- *employed labor:*

$$l_k = nu_k^\zeta \beta_k e^{-\zeta_k}; \quad (40)$$

- *nominal household's consumption:*

$$c_k^0 = \frac{\gamma_k e^{-\xi_k}}{\beta' \bar{L}' \gamma}; \quad (41)$$

- *prices over wage:*

$$\frac{p_k}{w} = \frac{e^{\xi_k}}{\mathbb{E}[e^{\rho_k}]}; \quad (42)$$

The actual quantities (y, z, c) obtain from the nominal ones, multiplying by the shocks e_k^ρ .

3 Data

This section describes the data used throughout the paper. Our objective is to construct a measure of climate-transition-risk exposure that is consistent with the theoretical framework developed in Section 2 and can be used both for the quantitative calibration and for the empirical analysis.

To this end, we assemble a novel panel combining U.S. firm-level financial information, sector-level carbon emissions, and detailed input-output linkages. Firm-level data are obtained from Compustat, carbon-emissions data from the Environmental Protection Agency (EPA), and production-network information from the Bureau of Economic Analysis (BEA) input-output tables. The resulting dataset covers the period 2012-2019.

The dataset serves two complementary purposes. First, it provides the empirical moments used in the calibration exercise of Section 4. Second, it allows us to test the model's central prediction that borrowing costs reflect not only firms' direct environmental exposure but also the transition risk transmitted through supply-chain relationships.

The remainder of this section describes the construction of the sample, the definition of borrowing-cost measures, and the methodology used to construct direct and network-adjusted emissions. A key contribution of the paper is indeed the construction of a model-consistent measure of network-adjusted carbon exposure. While direct emissions capture the carbon intensity of a sector's own production process, network-adjusted emissions additionally account for the emissions embodied in upstream suppliers. This measure mirrors the

network-amplified exposure that emerges in the theoretical model through the propagation of shocks across production networks.

3.1 Dataset Construction

Our empirical analysis is conducted at the firm year level using financial information from Compustat. Firm level variables include balance sheet items, capital structure, profitability, liquidity, and borrowing cost measures. Environmental exposure, however, is constructed at the six digit NAICS industry level, to avoid even unintentional misreporting of single firms. Each firm year observation is therefore assigned the direct and network adjusted CO₂ exposure of its corresponding NAICS6 industry.

The network based measure of environmental exposure is constructed using national input output tables from the Bureau of Economic Analysis, which capture production linkages across industries. By combining sector level emissions with these input output relationships, we obtain a measure of carbon exposure that reflects both firms’ own direct emissions and the emissions embodied in their upstream production networks. This approach allows us to capture transition risk transmitted along supply chains in a manner consistent with the theoretical framework.

Observations are included only if firms report a valid NAICS classification and non missing values for key financial variables, including interest expense, interest bearing debt, and total assets. To mitigate the influence of outliers driven by reporting issues or extreme accounting realizations, financial variables are trimmed at the top and bottom percentiles following standard practice. All trimming and winsorization procedures are applied consistently across variables and years.

Consistent with the focus of the theoretical model, we restrict the analysis to manufacturing industries, defined as two digit NAICS codes 31 to 33. Manufacturing sectors account for the majority of industrial emissions and exhibit substantial heterogeneity in both direct and network based carbon exposure. In contrast, service oriented industries display negligible direct emissions and limited upstream carbon intensity, making them less informative for identifying the role of production networks in the transmission of climate transition risk.

To compare outcomes across environmentally intensive and less intensive industries, we classify sectors as *brown* or *green* based on their relative carbon exposure within each year. Specifically, for every year, we consider the cross-industry distribution of lagged log carbon emissions and classify a sector as emission intensive (brown) if its lagged log emissions lie at or above the 80th percentile of the yearly distribution, while sectors below this threshold

are classified as green. Using a high-emission cutoff rather than the median ensures that the brown group captures industries with clearly elevated environmental exposure, thereby isolating the upper tail of the carbon distribution where transition risk is most likely to be economically salient. Using lagged emissions further guarantees that sector classification is predetermined with respect to contemporaneous financial outcomes.

This classification is implemented separately for direct emissions measured at the industry level and for network-adjusted emissions that capture exposure through input-output linkages. The resulting indicators therefore distinguish brown and green sectors either on the basis of their own lagged emissions or on the basis of their inherited exposure through production networks. Focusing on the upper tail of the distribution sharpens the contrast between high- and low-exposure sectors and reduces the risk that marginal emitters dilute the estimated differences.

Because the classification is time varying and constructed relative to the within-year cross-industry distribution, it allows for consistent comparisons of financial outcomes across sectors with differing degrees of environmental risk while abstracting from aggregate trends in emissions over time. In the preliminary descriptive statistics, however, we summarize outcomes by averaging over the full sample period to provide a first, unconditional comparison between brown and green sectors.

3.2 Measuring the Cost of Debt

Our main outcome variable is a firm level measure of the cost of debt, constructed as the ratio of interest expenses to interest bearing debt. For each firm i in year t , we define:

$$r_{i,t} = \frac{\text{xint}_{it}}{\text{dltt}_{it} + \text{dlc}_{it}} \times 100, \quad (43)$$

where xint_{it} denotes interest expense and $\text{dltt}_{it} + \text{dlc}_{it}$ corresponds to long term and short term interest bearing debt.⁴ This ratio captures the effective interest rate paid on all outstanding interest bearing obligations at the firm level, aggregating across maturities and instruments.⁵

⁴In the Appendix we also build a measure based on total liabilities.

⁵To mitigate the influence of implausible values driven by reporting issues or timing mismatches between interest expenses and debt stocks, we apply several data quality filters. We drop observations with missing NAICS codes or missing values for interest expense or interest-bearing debt. We winsorize key balance sheet variables, including xint , dltt , dlc , lt , and at , at the 2.5th and 97.5th percentiles. Based on these cleaned components, we compute the implied interest rate and initially restrict the sample to observations with interest rates less than or equal to 30 percent. We then winsorize the resulting interest rate measure at the

This firm level measure of the cost of debt is widely used in empirical corporate finance as a proxy for borrowing costs, not only when market yields are unavailable, but also when the objective is to capture the effective interest rate paid on a firm’s entire stock of outstanding debt, as in [Pittman and Fortin \(2003\)](#). By construction, this measure aggregates interest expenses across short and long term obligations and relates them to the corresponding debt stocks, providing a comprehensive summary of firms’ realized financing costs.

In our context, this choice is particularly appropriate because it aligns closely with the theoretical mechanism studied in the paper. In the model, interest rates are set by banks as equilibrium objects that reflect expected default risk on the full liability structure of the firm, rather than on a single debt instrument or maturity. The ratio of interest expenses to interest bearing debt therefore captures the relevant object priced by lenders when assessing transition risk as it materializes.

Moreover, focusing on the effective interest rate allows us to account for heterogeneity in debt maturity, contractual terms, and refinancing decisions that may themselves respond to climate policy expectations. Unlike quoted loan spreads or bond yields, which pertain to specific instruments at issuance, the realized cost of debt reflects the cumulative impact of climate related risk pricing across the firm’s entire balance sheet. This makes it particularly well suited to studying how transition risk, both direct and network based, is internalized by credit markets in equilibrium.

3.3 Measuring Network-Adjusted Emissions

A central contribution of this paper is the construction of a network-based measure of carbon exposure that captures how transition risk propagates through production linkages across industries. To this end, we distinguish between direct emissions, also known as *scope 1*, and our novel measure of network-adjusted emissions, both defined at the six-digit NAICS level. While direct CO₂ emissions measure the carbon directly generated by a sector’s own production activities, they fail to account for the substantial indirect exposure arising from reliance on upstream inputs produced in carbon-intensive industries.

Our network-adjusted emissions measure explicitly incorporates inter-industry dependencies and weights upstream emissions by their economic relevance in each sector’s production

5th and 95th percentiles, following the approach in [Pittman and Fortin \(2003\)](#). The same 30 percent cutoff and 5 percent winsorization are applied to the alternative interest rate measure based on total liabilities. Finally, we winsorize leverage at the 2.5th and 97.5th percentiles to reduce the influence of extreme capital structure realizations.

structure, in line with the theoretical framework developed in this paper.⁶ This approach recognizes that sectors may face heightened transition risk even in the absence of their own direct emissions, simply due to their position within carbon-intensive supply chains.

We construct network-adjusted CO₂ emissions sector-by-sector by combining observed six-digit NAICS sectoral direct emissions with input-output relationships across industries. Let CO2_t denote the vector of observed sectoral direct emissions at time t , and let $\mathbf{L}_t$ denote the corresponding Leontief inverse matrix capturing both direct and indirect production linkages across sectors. Consistent with the calibration of the model⁷, network-adjusted emissions for sector k are then computed as:

$$\text{Network CO2}_{k,t} = \sum_{j \in \nu} \mathbf{L}_{jk,t} \cdot \text{CO2}_{j,t} = \sum_{j \in \nu} \sum_{h=0}^{+\infty} (\mathbf{A}'_{jk})^h \cdot \text{CO2}_{j,t} \quad \forall i \in k \quad (44)$$

This construction propagates direct emissions throughout the entire production network, weighting upstream carbon intensity by each sector’s dependence on intermediate inputs while allowing financial conditions to amplify exposure. As a result, network-adjusted CO₂ captures the total embodied emissions associated with each sector’s output, reflecting both direct emissions and those inherited through supply-chain linkages.

By explicitly accounting for the structure of production networks, this measure provides a more accurate representation of firms’ effective carbon exposure and the channels through which transition risk is transmitted across industries. It is particularly relevant in carbon-intensive economies, where upstream shocks may generate substantial downstream financial consequences.

Figure 1 illustrates the distribution of both CO2 and **Network CO2** in log scale, where unit of measure is expressed in million metric tons of CO₂ equivalent (MtCO₂e), calculated using 100-year global warming potentials from the IPCC Fourth Assessment Report (AR4). As discussed above, restricting the analysis to the manufacturing sector allows us to isolate emissions dynamics in sectors with substantial and economically relevant carbon intensity. The distribution of direct emissions is strongly right-skewed, with a large concentration of sectors exhibiting low emission levels and a long right tail driven by a small number of highly polluting industries.

In contrast, the network-adjusted measure substantially reweights emissions across sectors by accounting for input–output linkages. Sectors that appear to generate negligible

⁶Details about the Input-Output matrices and subsequent derivations are available in Appendix C.1.3.

⁷In Appendix B we show that the $\mathbf{A}$ matrix need to be scaled by $(1 + r_t)$, where r_t represents the sector-specific output interest rate.

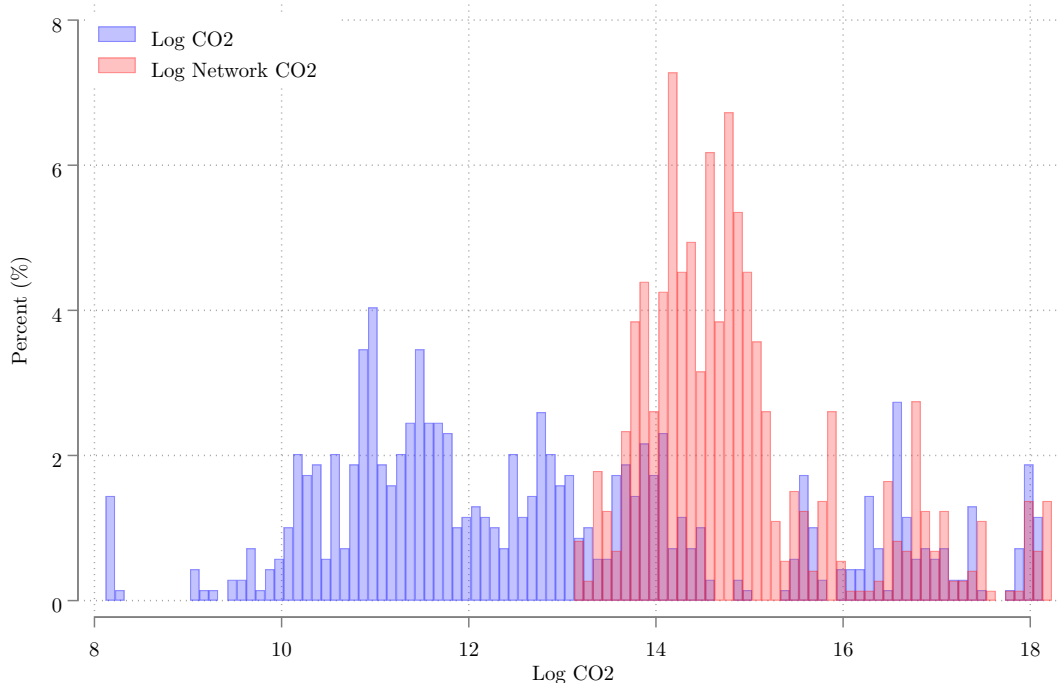


Figure 1: Histograms of CO2 and Network CO2 Emissions

Notes: This figure displays histograms of firm-level CO2 (blue bins) emissions and Network CO2 (red bins) emissions, both expressed in logarithmic scale. Each histogram is based on approximately 6,000 observations. The y-axis reports the relative frequency of observations, expressed as percentages. CO2 measures direct emissions at the firm level, while Network CO2 captures emissions associated with firms' production networks as defined in this section.

direct emissions exhibit positive network-adjusted emissions due to their reliance on carbon-intensive upstream suppliers. This redistribution highlights how embodied emissions propagate through production networks, effectively assigning carbon exposure to downstream sectors according to their economic dependence on polluting industries.

3.4 Descriptive Statistics

Table 1 reports descriptive statistics for the main variables used in the empirical analysis. The sample covers the period from 2012 to 2019, providing balanced coverage around the entry into force of the Paris Agreement, which will be part of this empirical analysis.

The firm-level cost of debt, measured as interest expenses relative to interest-bearing debt, has an average of 6.18 percent and a standard deviation of 5.07 percentage points. The distribution exhibits moderate right skewness, with values ranging from close to zero to

Table 1: Descriptive Statistics

	Average	SD	Min	Max	Skewness	Count
Interest Rate (Debt)	6.18	5.07	0.00	23.03	0.96	6965
C02	4.26	16.96	0.00	182.51	7.59	6986
Network C02	6.35	17.85	0.45	190.91	7.38	6982
Log Z	0.40	1.12	-6.35	2.18	-1.15	1579

Notes: This table reports descriptive statistics for firm-year observations. Borrowing costs are measured as interest expenses relative to interest-bearing debt or total liabilities. Direct emissions (**C02**) and network-adjusted emissions (**Network C02**) are constructed at the six-digit NAICS industry level and expressed in million metric tons of CO₂ equivalent (MtCO₂e), using 100-year global warming potentials from the IPCC Fourth Assessment Report (AR4). Emissions measures are assigned to firms based on industry affiliation.

just above 23 percent. As discussed in Section 5, extreme values of interest rates are trimmed to mitigate the influence of reporting issues and timing mismatches between interest expenses and debt stocks.

Carbon emissions display substantial heterogeneity across industries. Direct emissions, measured by **C02** and expressed in million metric tons of CO₂ equivalent (MtCO₂e), have an average of 4.26 and a standard deviation of 16.96, with a highly right-skewed distribution and a maximum exceeding 182 MtCO₂e. Network-adjusted emissions exhibit a higher average of 6.35 and similarly large dispersion, with even stronger right skewness. This reflects the amplification of carbon exposure through upstream production linkages, whereby sectors with low or moderate direct emissions inherit substantial embodied emissions from carbon-intensive suppliers.

The number of observations for emissions variables is lower than for financial variables because emissions are observed at the sector-year level rather than the firm-year level. In particular, log emissions are available for 636 sector-year observations for direct emissions and 658 for network-adjusted emissions, reflecting the number of six-digit NAICS manufacturing sectors with non-missing emissions data in each year of the sample period. These sector-level measures are then mapped to firms operating within the corresponding industries.

Firm-level productivity ⁸, measured by the logarithm of total factor productivity, $\log Z_{i,t}$, exhibits considerable dispersion, with an average of 0.40 and a standard deviation of 1.12. The distribution is markedly left-skewed, with a minimum of -6.35 , indicating the presence of a subset of firm-year observations characterized by persistently low productivity levels.

⁸A more detailed section on the derivation of TFP is available in Appendix C.2.2.

At the same time, the upper tail reaches values above 2, reflecting substantial heterogeneity in productive performance across firms.

Overall, the descriptive statistics highlight pronounced variation in borrowing costs, environmental exposure, and productivity across firms and sectors. Emissions are highly concentrated among a small subset of industries, while network-adjusted emissions redistribute carbon exposure across downstream sectors. These patterns motivate the subsequent analysis, which examines how both direct and network-based carbon exposure shape the pricing of corporate debt.

4 Quantitative Analysis and Calibration

In this section, we quantitatively illustrate the mechanism developed in the theoretical model before turning to the empirical analysis. The objective is to evaluate whether the equilibrium framework introduced in Section 2 can generate realistic differences in borrowing costs between green and brown sectors once production-network linkages and endogenous default risk are taken into account.

Using the data described in Section 3, we assess the quantitative relevance of this mechanism. In particular, we calibrate the primitive shock distribution so that the model reproduces the average borrowing-cost spread observed between green and brown sectors. The calibration is intentionally parsimonious: rather than assigning sector-specific shock distributions, we assume that all green sectors share one common shock intensity and all brown sectors share another.

The quantitative exercise serves two purposes. First, it provides a numerical illustration of the mechanism developed in Section 2. Second, it quantifies the extent to which relatively small differences in primitive environmental risk can generate meaningful differences in borrowing costs once shocks propagate through production networks.

The calibration allows us to assess whether the interaction between environmental shocks, network propagation, and endogenous default risk is sufficient to account for the borrowing-cost differentials observed in the data. More broadly, it provides a bridge between the theoretical framework and the empirical analysis.

The results show that network amplification plays an important role in translating differences in environmental exposure into financing conditions. Higher borrowing costs do not arise from an exogenously imposed premium on brown sectors, but emerge endogenously through the interaction between environmental risk, production-network propagation, and

competitive bank pricing.

Calibration. For each sector i , the primitive downside shock is specified as

$$X_i \sim \text{Exp}(\lambda_i), \quad \eta_i = -X_i \leq 0. \quad (45)$$

Since

$$\mathbb{E}[X_i] = \frac{1}{\lambda_i}, \quad (46)$$

a lower value of λ_i corresponds to a larger expected downside shock. If brown sectors are more exposed to transition risk, the calibration should therefore imply

$$\lambda_b < \lambda_g. \quad (47)$$

We impose a parsimonious two-parameter structure:

$$\lambda_i = \begin{cases} \lambda_g, & i \leq n_1, \\ \lambda_b, & i > n_1, \end{cases} \quad (48)$$

Sectors are classified using the median of network-adjusted emissions, so that the calibration is based on the same network-exposure concept used throughout the empirical analysis.

The calibration is implemented numerically using Monte Carlo simulation. For each candidate parameter pair (λ_g, λ_b) , the algorithm first draws primitive shocks at the sector level from the corresponding exponential distributions. These primitive shocks are then propagated through the empirical production network using the Leontief inverse, generating the network-adjusted shocks that determine equilibrium repayment. Given these repayment outcomes, the model solves the competitive bank zero-profit condition sector by sector and computes the associated equilibrium borrowing costs implied by the model, $\bar{r}_g^{\text{model}}$ and $\bar{r}_b^{\text{model}}$. The calibrated parameters are the values of (λ_g, λ_b) that minimize the distance between the model-implied and empirically observed green and brown average borrowing costs, $\bar{r}_g^{\text{obs}}$ and $\bar{r}_b^{\text{obs}}$, where the average is taken over firms and time. ⁹

$$(\hat{\lambda}_g, \hat{\lambda}_b) = \arg \min_{\lambda_g > 0, \lambda_b > 0} [\bar{r}_g^{\text{model}} - \bar{r}_g^{\text{obs}}]^2 + [\bar{r}_b^{\text{model}} - \bar{r}_b^{\text{obs}}]^2. \quad (49)$$

The calibration targets the observed average borrowing costs of green and brown sectors

⁹A detailed step-by-step description of the calibration procedure is provided in Appendix B.

using the two-parameter specification described above.

Quantitatively, the calibration yields

$$\hat{\lambda}_b \approx 5.45, \quad \hat{\lambda}_g \approx 5.74. \quad (50)$$

The calibrated parameters satisfy

$$\hat{\lambda}_b < \hat{\lambda}_g, \quad (51)$$

implying that brown sectors are calibrated to experience larger expected primitive downside shocks than green sectors.

The implied expected primitive downside shock is therefore approximately

$$\mathbb{E}[X_b] \approx \frac{1}{5.45} \approx 0.183, \quad (52)$$

for brown sectors and

$$\mathbb{E}[X_g] \approx \frac{1}{5.74} \approx 0.174, \quad (53)$$

for green sectors. Also, the calibration implies that brown sectors are exposed to systematically larger downside risk, in the sense of first order stochastic dominance,¹⁰ although the quantitative difference in primitive shock intensity remains relatively modest.

Despite this relatively small primitive difference, the model generates a meaningful equilibrium borrowing-cost spread once shocks propagate through the production network. In the data, the average observed borrowing rate is approximately

$$\bar{r}_g^{\text{obs}} \approx 0.062, \quad \bar{r}_b^{\text{obs}} \approx 0.066, \quad (54)$$

implying an observed brown-green spread of roughly 36 basis points. The calibrated model reproduces these moments almost exactly.

The calibration therefore illustrates the central mechanism of the model: relatively modest differences in primitive environmental downside risk can generate economically meaningful differences in equilibrium financing conditions once network amplification and endogenous bank pricing are taken into account.

Quantitative implications. Total shocks can be computed analytically for each sector as in equation 34 and 35. Each green and brown sector is subject to a different total shock and

¹⁰This happens because the exponential distribution is monotone in its parameter.

they are not independent. This is due to the fact that they mirror the complex relationships among sectors. This affects the equilibrium values of both the financial and the real variables of the model.

Financial equilibrium quantities can be computed analytically using the definitions, while real variables obtain from Theorem 3. Table 2 reports the main financial equilibrium variables. It shows that brown sectors have a higher equilibrium debt costs, with $\Delta\zeta = 0.004$, and larger network-adjusted total cost of debt, with $\Delta\xi = 0.015$.

Table 2: Financial Variables by Green and Brown Sectors

Comparison	λ	ζ	ξ	ν	ν^ζ
Brown - Green	-0.36	0.004	0.015	0	0.003

Notes: This table reports equilibrium financial variables. The parameter λ denotes the calibrated downside-shock intensity. The variable ζ is the equilibrium debt-cost variable implied by the bank zero-profit condition, ξ is the total cost of debt, and ν and ν^ζ denote baseline and levered Bonacich centrality. Results are computed using one million Monte Carlo simulations.

The fact that $\Delta\xi > \Delta\zeta$ indicates that the transition-risk component is amplified through the production network and priced by banks in equilibrium. Considering leverage, network centrality is also higher for brown sectors, with $\Delta\nu^\zeta = 0.003$.

Table 3 reports real equilibrium variables. It shows that brown sectors have slightly higher equilibrium output and labor allocations, with $\Delta y = 0.002$ and $\Delta l = 0.002$. This happens because, on average, their greater cost of debt (total or not) does not compensate their higher levered centrality. Indeed, both output and labor are increasing in the centrality of the sector, once leverage is taken into account in computing it, and decreasing in the cost of debt, as in Theorem 2. Consumption is essentially unchanged. The largest difference appears in relative prices: the brown-minus-green differential in the price-wage ratio is $\Delta(p/w) = 0.083$. Prices are indeed increasing in the total cost of debt and decreasing in the expected total shocks. The fact that, on average, in the data green prices are below brown ones means that the cost of debt effect dominates the expected shock one. Table 2, indeed, shows that not only the average cost of debt ζ is smaller for green than for brown sectors, but also that the discrepancy increases when we consider the total cost, ξ . This produces the dominant effect in prices. Obviously, the table summarizes a more nuanced behavior of specific sector, since we report only the average parameters.

To give a sense of the richness of the actual US manufacturing economy behind, Figure 2 reports the distribution of the model-implied debt-cost variable ζ across green and brown

Table 3: Equilibrium Quantities by Green and Brown Sectors

Comparison	y	l	c	p/w
Brown - Green	0.002	0.002	-0	0.083

Notes: This table reports equilibrium quantities and relative prices. The variables reported are equilibrium output y , labor demand l , consumption c , and the equilibrium price-wage ratio p/w . Results are computed using one million Monte Carlo simulations.

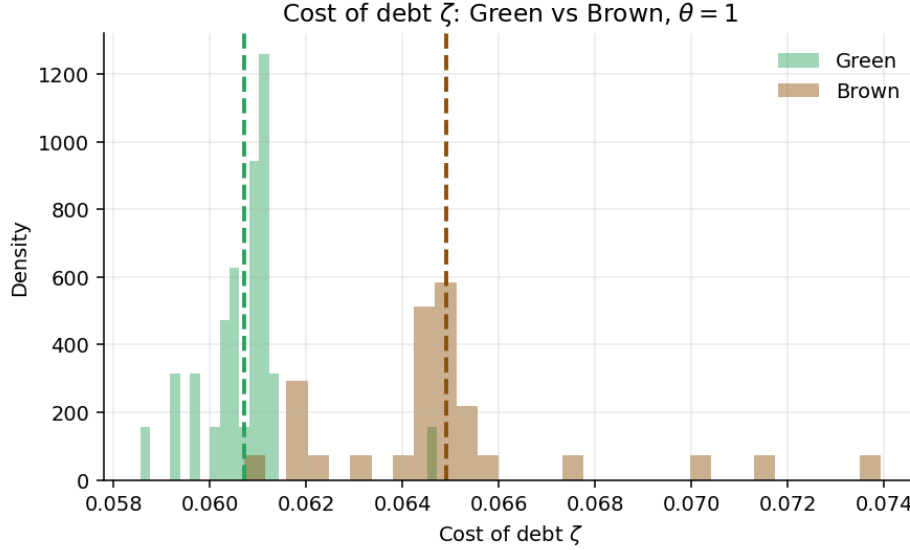


Figure 2: Distribution of the model-implied debt-cost variable ζ for green and brown sectors.

Notes: Histograms of the model-implied debt-cost variable ζ for green and brown sectors for the calibrated economy. Dashed vertical lines indicate group means.

sectors. The histogram for brown sectors is shifted to the right relative to green sectors, implying the higher equilibrium average of Table 2. The overlap between the two distributions reflects the fact that the green-brown network exposure are not completely separated, as corollary 1 would require. Each green sector, if it were isolated or linked only to green would require a lower cost of debt than any brown. Interconnectedness of the two groups make the overlap appear.

Table 4 shows that average firm profits are essentially unchanged across groups, while the dispersion of firm profits is slightly larger for brown sectors. More importantly, the firm-default probability is higher for brown sectors by 0.699 percentage points. This difference reflects the additional transition-risk exposure of brown sectors and its amplification through the production network. Brown sectors have more risky profits, also when captured by a

Table 4: Profit and Default Moments by Green and Brown Sectors

Comparison	Firms Mean	Firms SD	Firm Default (%)
Brown - Green	0	0.001	0.699

Notes: This table reports simulated firm-profit distributions and default moments. The table reports average firm profits, the standard deviation of firm profits, and the probability of firm default. Default probabilities are computed from one million Monte Carlo simulations of the calibrated equilibrium economy under $\theta = 1$.

simple standard deviation.

Overall, the quantitative exercise provides strong support for the central mechanism of the model. Brown sectors are calibrated to face larger expected transition-related downside shocks and, as a consequence, exhibit systematically higher equilibrium debt costs, larger network-adjusted financial distortions, and greater default risk. Importantly, these differences do not arise from exogenously imposed financing premia. Rather, they emerge endogenously from the interaction between environmental risk, production-network propagation, and competitive bank pricing. Although the calibrated difference in primitive shock exposure between brown and green sectors is relatively modest, the production network amplifies these shocks through upstream and downstream linkages, generating economically meaningful differences in equilibrium outcomes. The quantitative exercise therefore illustrates how climate transition risk can be magnified through supply-chain connections and ultimately translated into higher borrowing costs, prices and greater financial fragility in equilibrium.

5 Empirical Validation

This section provides an empirical validation of the theoretical framework developed in Section 2. The model predicts that, in equilibrium, corporate borrowing costs reflect firms' exposure to climate transition risk, where such exposure extends beyond firms' own direct emissions and is transmitted through production networks. As a result, lenders price network adjusted carbon exposure rather than direct sectoral emissions alone. In Appendix A.2.1 we verify that indeed the US manufacturing network meets the conditions in Theorem 2.

5.1 Brown and Green Sectors in the Data

Table 5 reports summary statistics for productivity across green and brown sectors. Green sectors have an average log productivity of 0.37, with a standard deviation of 1.11, a minimum

Table 5: **Network** CO2 Emissions and Productivity

	Average	SD	Min	Max	Skewness	Count
Green	0.37	1.11	-6.35	2.18	-1.13	1371
Brown	0.63	1.14	-4.60	2.16	-1.36	208
Total	0.40	1.12	-6.35	2.18	-1.15	1579

Notes: This table reports summary statistics for firm-level productivity across green and brown manufacturing sectors. Sector classification is based on network-adjusted carbon emissions measured at the six-digit NAICS industry level. In each year t , a sector is defined as brown if its lagged log **Network** CO2 emissions are at or above the 80th percentile of the cross-industry distribution, and as green otherwise. Productivity is measured as the log of total factor productivity, $Z_{i,t}$, constructed using a standard Solow residual.

value of -6.35, and a maximum of 2.18. Brown sectors exhibit a higher mean productivity of 0.63 and a slightly larger dispersion, with a standard deviation of 1.14, a minimum of -4.60, and a maximum of 2.16. The brown group represents the upper tail of the emissions distribution and comprises 208 observations, compared with 1,371 observations for green sectors, for a total sample size of 1,579 observations.

Both groups display negatively skewed productivity distributions, with skewness equal to -1.13 for green sectors and -1.36 for brown sectors, indicating exposure to low productivity realizations. The skewness is slightly more pronounced among brown sectors, suggesting a thicker lower tail despite their higher mean productivity. At the same time, the upper tails of the distributions are similar across the two groups, with nearly identical maximum values.

Overall, the descriptive evidence indicates that sectors located in the upper tail of the carbon exposure distribution combine higher average productivity with substantial dispersion in outcomes. These patterns motivate the subsequent empirical analysis investigating how environmental exposure influences credit market conditions and propagates through production networks.

Consistent with our theoretical framework, which allows for higher productivity in emission-intensive sectors, we provide this descriptive statistics table to highlight empirical patterns that are well grounded in the existing literature. In particular, the observed productivity advantage of high-emission industries aligns with the argument that environmental constraints interact with firms' technological choices and competitive strategies, as emphasized by [Porter \(1991\)](#) and [Porter and van der Linde \(1995\)](#), who stress that productivity and competitiveness need not be adversely affected by environmental regulation. Moreover, our findings are consistent with evidence from the European context showing that

environmental policies can coexist with, and in some cases even enhance, economic performance and productivity outcomes, as documented by [Dechezlepretre et al. \(2021\)](#). Taken together, these descriptive statistics validate our modeling assumptions and motivate the subsequent empirical analysis of how environmental exposure shapes credit market conditions and propagates through production networks.

5.2 Direct vs Network-Adjusted Emissions

We examine how exposure to direct and network-based carbon emissions affects corporate borrowing costs using a panel regression framework with high-dimensional fixed effects. The dependent variable is the firm-level effective interest rate paid by firm i in year t . Carbon exposure variables are constructed at the six-digit NAICS industry level and assigned to firms operating within those sectors. The explanatory variables are expressed in logarithms, so coefficients measure semi-elasticities: they capture the percentage-point change in the interest rate associated with a one percent change in carbon exposure.

Our baseline specification relates firm-level borrowing costs to lagged industry-level emissions, with $\mathcal{C} = \{\text{C02}, \text{Network C02}\}$, and takes the form

$$r_{i,t} = \alpha + \beta_1 \log \mathcal{C}_{k(i),t-1} + \beta_2 X_{i,t-1} + \gamma_i + \delta_t + \varepsilon_{i,t}, \quad (55)$$

where $k(i)$ denotes the six-digit NAICS industry in which firm i operates. The vector $X_{i,t-1}$ includes firm-level controls that affect borrowing costs, specifically log leverage and log firm size. All specifications absorb firm fixed effects γ_i , which control for time-invariant firm characteristics, and year fixed effects δ_t , which capture aggregate macroeconomic conditions, credit market cycles, and common regulatory developments. Standard errors are clustered at the firm level to account for serial correlation within firms over time. For brevity, the coefficients on control variables and fixed effects are included in the regressions but not reported in the table.

Table 6 reports the main regression results across three specifications. Column (1) relates firm-level borrowing costs to lagged direct emissions at the industry level. The coefficient on $\log \text{C02}$ is 2.434 and statistically significant at the 10 percent level. This estimate implies that a one percent increase in industry-level direct emissions is associated with an increase of approximately 2.4 basis points in firms' borrowing costs, conditional on firm characteristics and aggregate time effects.

Column (2) replaces direct emissions with the network-adjusted emissions measure. The

Table 6: Direct vs Network-Adjusted Emissions

	Interest Rate (1)	Interest Rate (2)	Interest Rate (3)
Log C02 _{t-1}	2.434* (0.910)		2.792** (0.952)
Log Network C02 _{t-1}		2.546** (0.890)	
Log Network C02 _{t-1}			1.922* (0.883)
Firm FEs	Yes	Yes	Yes
Year Fes	Yes	Yes	Yes
Sectors SEs	Yes	Yes	Yes
R-squared	0.686	0.682	0.686
N	1744	1778	1744

Notes: This table reports estimates from the baseline regression model in Equation 55. The dependent variable is the firm-level interest rate. Column (1) includes lagged direct industry-level C02 emissions. Column (2) replaces direct emissions with lagged network-adjusted emissions (**Network C02**). Column (3) includes lagged direct emissions and the residual component of network-adjusted emissions orthogonalized with respect to direct emissions. All specifications include log leverage and log firm size as controls, as well as firm and year fixed effects. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

estimated coefficient on log **Network C02** is 2.546 and statistically significant at the 5 percent level. This result indicates that firms operating in sectors embedded in more carbon-intensive production networks face significantly higher borrowing costs, even after controlling for leverage, size, firm fixed effects, and year fixed effects.

Column (3) includes lagged direct emissions together with the residual component of network-adjusted emissions obtained by orthogonalizing **Network C02** with respect to direct C02. This residualized measure isolates variation in network exposure that is not mechanically driven by a sector's own emissions. In this specification, the coefficient on log C02 increases to 2.792 and remains statistically significant at the 5 percent level, while the coefficient on the residualized network measure is 1.922 and statistically significant at the 10 percent level.

These findings indicate that both direct emissions and the network-based component of carbon exposure are priced in credit markets. Importantly, the residualized specification demonstrates that network exposure exerts an independent effect on borrowing costs beyond firms' own direct emissions. The magnitude of the network coefficient remains economically meaningful, suggesting that creditors internalize transition risk transmitted through supply-

chain linkages rather than focusing exclusively on firms’ direct carbon footprint.

Taken together, the results in Table 6 highlight the importance of accounting for network-based carbon exposure when assessing the pricing of corporate credit. Once leverage, firm size, firm heterogeneity, and aggregate time effects are fully absorbed through controls and fixed effects, exposure to carbon-intensive production networks continues to exert a robust and economically meaningful impact on borrowing costs. These findings are consistent with a view of climate transition risk as a systemic phenomenon embedded in production networks.

5.3 Causality and the Paris Agreement: A Network Perspective

In this subsection, we move beyond correlations and investigate the causal impact of climate policy expectations on firms’ borrowing costs. Our analysis focuses specifically on the network-adjusted emissions measure, which constitutes the central object of interest in this study. In the previous sections, we document that firms’ exposure to carbon-intensive production networks plays a statistically and economically significant role in explaining the cross-sectional variation in interest rates charged by banks. Having established the relevance of the network channel, we now examine whether this relationship reflects a causal effect that becomes particularly salient in the presence of a major climate policy shock.

To identify causality, we exploit the entry into force of the Paris Agreement in late 2016 as a global, externally imposed shock to climate governance. The Paris Agreement marked a coordinated international commitment to long-run decarbonization and signaled a structural shift in expectations regarding the future stringency of environmental regulation across countries and industries. Crucially, this shift was broad-based and externally determined, rather than driven by firm-level financial conditions or sector-specific credit dynamics.

Importantly for identification, the Paris framework applied to all major emitting economies, including the United States. Although adopted in December 2015, the agreement formally entered into force in November 2016 following ratification by a critical mass of large emitters. Under the Obama administration, the United States formally joined the agreement, signaling a credible commitment to nationally determined contributions and to a trajectory of progressively tighter climate regulation. While the Trump administration announced its intention to withdraw in June 2017, institutional treaty rules implied that the United States remained a party until November 2020. The subsequent reentry under the Biden administration in early 2021 further reinforced the perception that international climate policy reflects a persistent and structural shift rather than a transitory political stance. Our sample period spans this entire phase of activation and implementation, making

the Paris Agreement a suitable quasi-natural experiment.

We exploit this institutional setting using two complementary empirical strategies tailored to the network exposure measure. First, we implement a difference-in-differences design comparing the evolution of borrowing costs between firms more exposed to carbon-intensive production networks (“network-brown”) and those with lower network exposure (“network-green”) before and after the Paris Agreement. Second, we complement this analysis with an event-study approach that traces the dynamic adjustment of credit markets around the policy event and allows us to formally assess the parallel trends assumption underlying the difference-in-differences framework.

Focusing on the network-adjusted measure is conceptually central to our research question. If financial markets internalize transition risk not only through firms’ own emissions but also through their embeddedness in carbon-intensive supply chains, then the repricing of credit risk should materialize most clearly along the network dimension following a credible climate policy shock. The Paris Agreement provides precisely such a shock: it generated a common and plausibly exogenous shift in expectations about future regulatory stringency, thereby increasing the salience of indirect carbon exposure transmitted through production networks.¹¹

By leveraging this quasi-natural experiment, we directly assess whether firms’ exposure to carbon-intensive production networks causally affects their cost of debt financing. In doing so, we isolate the network channel as the key mechanism through which climate transition risk is transmitted to credit markets.

Difference-in-Differences Approach (Network Exposure). We now implement a difference-in-differences design that exploits cross-industry heterogeneity in *network-adjusted* carbon exposure and compares firm-level borrowing costs before and after the entry into force of the Paris Agreement. Consistent with the core objective of this study, treatment status is defined exclusively using the network-based emissions measure.

Industries are classified as network-brown if their lagged growth in log network-adjusted emissions lies above the 80th percentile of the cross-industry distribution within a given year. In other words, the treatment indicator equals one for industries in the top 20 percent

¹¹A growing literature exploits the entry into force of the Paris Agreement as a quasi-natural experiment to identify the causal effects of climate policy expectations on economic and financial outcomes. Prior work documents improvements in global environmental efficiency using a fuzzy regression discontinuity design (Salman et al., 2022) and positive stock market responses for firms exposed to green revenue opportunities (Kruse et al., 2024). These findings support the interpretation of the Paris Agreement as an externally imposed shift in expectations about future climate policy.

of the distribution of network carbon growth and zero otherwise. This high-exposure cutoff ensures that the treated group captures sectors with pronounced increases in inherited carbon intensity, thereby isolating industries most likely to face elevated transition risk. Firms operating in these network-brown industries constitute the treated group, while firms in network-green industries form the control group.

The identifying assumption is that, absent the Paris Agreement, borrowing costs for firms in network-brown and network-green industries would have followed parallel trends. The Paris Agreement represents a common and externally imposed policy shock, and identification comes from differential exposure to transition risk through production networks.

We estimate the following two-way fixed effects specification:

$$r_{i,t} = \alpha + \delta (\text{Network } B_{k(i)} \times \text{Post}_t) + \mu_i + \lambda_t + \varepsilon_{i,t}, \quad (56)$$

where $r_{i,t}$ denotes the firm-level interest rate, **Network** $B_{k(i)}$ indicates whether firm i operates in a network-brown industry (top 20 percent of network emission growth), and Post_t equals one for years 2017 and later.

Firm fixed effects μ_i control for time-invariant firm characteristics such as baseline credit quality, business model, or managerial practices. Year fixed effects λ_t absorb aggregate macroeconomic conditions, credit supply shocks, and common regulatory developments affecting all firms. Including both firm and year fixed effects is essential in a multi-year difference-in-differences design. Firm fixed effects remove permanent cross-sectional differences between treated and control firms, while year fixed effects absorb common time shocks. Identification therefore relies solely on within-firm changes in borrowing costs relative to the contemporaneous evolution of the control group. Standard errors are clustered at the firm level to account for serial correlation.

Importantly, we do not include additional firm-level controls in the baseline DiD specification. Because the Paris Agreement constitutes an externally imposed policy shock, and because firm and year fixed effects already absorb time-invariant heterogeneity and aggregate trends, the interaction term isolates the differential post-2016 change in borrowing costs attributable to network-based carbon exposure.¹²

Table 7 reports the estimates. Column (1), which includes firm fixed effects but not year fixed effects, yields a coefficient of 0.868, statistically significant at the 1 percent level.

¹²Prior to constructing the treatment indicator, we winsorize emissions and log emissions at the 1st and 99th percentiles and trim extreme values to mitigate the influence of outliers. Network-brown status is defined using the 80th percentile of the annual cross-industry distribution of lagged log changes in network-adjusted emissions, consistent with the empirical construction described in the data section.

Table 7: Difference-in-Differences Estimates: Paris Agreement and Network Exposure

	Interest Rate (1)	Interest Rate (2)
Network B x T	0.868*** (0.261)	0.795** (0.280)
Firm FEs	Yes	Yes
Year FEs	No	Yes
Sectors SEs	Yes	Yes
R-squared	0.597	0.607
N	3250	3250

Notes: This table reports difference-in-differences estimates of the effect of the Paris Agreement on firm-level borrowing costs. Treatment status is defined using network-adjusted carbon exposure at the six-digit NAICS level. Industries are classified as brown if their lagged growth in log network-adjusted emissions lies above the 80th percentile of the annual cross-industry distribution. The post indicator equals one for years 2017 and later. Column (1) includes firm fixed effects only. Column (2) includes both firm and year fixed effects. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

Column (2) adds year fixed effects, thereby controlling for aggregate credit market conditions and common macroeconomic shocks. The coefficient remains positive and statistically significant, with an estimate of 0.795. The stability of the estimate across specifications indicates that the differential increase in borrowing costs is not driven by time-invariant firm characteristics or common time effects, but instead reflects a relative repricing of firms exposed to carbon-intensive production networks.

Overall, the difference-in-differences results provide causal evidence that the Paris Agreement triggered a differential increase in borrowing costs for firms embedded in carbon-intensive supply chains. Consistent with the theoretical framework, credit markets internalize transition risk primarily through network-based exposure, particularly for industries located in the upper tail of the network carbon growth distribution.

Event Study Approach. To complement the difference in differences analysis and to examine the dynamic adjustment of borrowing costs around the Paris Agreement, we estimate an event study specification centered on the policy implementation year 2017, with 2016 serving as the omitted baseline year. While the DiD framework identifies an average post treatment effect, the event study design allows us to assess the credibility of the parallel

trends assumption and to trace the timing and persistence of credit market responses to climate policy expectations.

Treatment status is defined using the same network based classification employed in the DiD analysis. Industries are classified as network brown if their lagged growth in log network adjusted emissions lies at or above the 80th percentile of the annual cross industry distribution. Firms operating in these industries are assigned a treatment indicator equal to one, while firms in the remaining industries form the network green control group. By focusing on the top 20 percent of the distribution, this classification isolates industries experiencing pronounced increases in inherited carbon exposure, thereby concentrating on sectors most plausibly exposed to elevated transition risk following a credible climate policy shock.

We estimate the following two way fixed effects event study specification:

$$r_{i,t} = \alpha + \sum_{s \neq -1} \beta_s (\text{Network } B_{k(i)} \times 1\{t - 2017 = s\}) + \gamma_i + \lambda_t + \varepsilon_{i,t}, \quad (57)$$

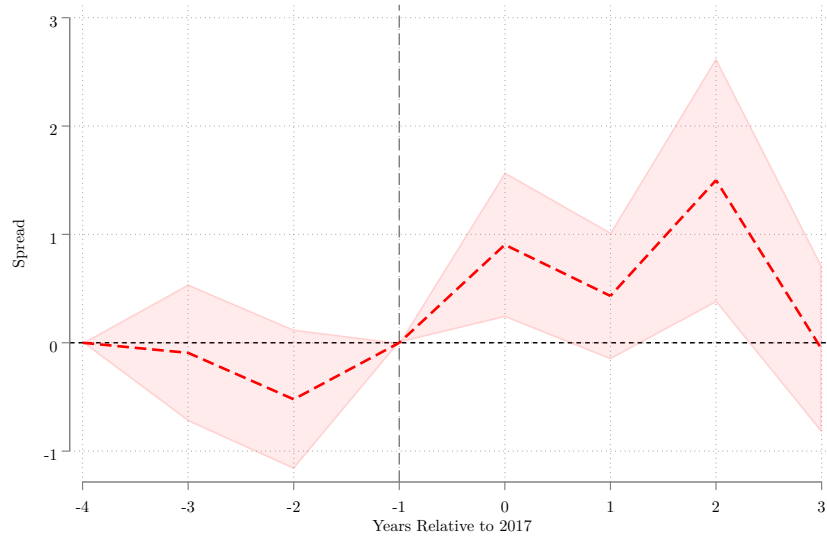
where $r_{i,t}$ denotes the firm level effective interest rate, **Network** $B_{k(i)}$ is the network brown indicator for firm i 's six digit NAICS industry $k(i)$, and $1\{t - 2017 = s\}$ is an event time dummy equal to one in relative year s . The year immediately preceding implementation, $s = -1$, corresponding to 2016, is omitted from the regression, so each coefficient β_s measures the differential borrowing cost between network brown and network green firms in relative year s compared to the pre Agreement baseline.

Firm fixed effects γ_i absorb all time invariant firm characteristics, including persistent differences in credit quality, capital structure, and business models. Year fixed effects λ_t control for aggregate macroeconomic conditions, common credit supply shocks, monetary policy changes, and other regulatory developments affecting all firms simultaneously. Standard errors are clustered at the firm level to allow for serial correlation in borrowing costs over time.

The pre treatment coefficients, that is for $s < 0$, provide a direct test of the parallel trends assumption. Under the identifying assumption, borrowing costs for network brown and network green firms should evolve similarly prior to the Paris Agreement, implying that these coefficients should be close to zero and statistically insignificant. Post treatment coefficients, that is for $s \geq 0$, capture the dynamic repricing of transition risk following the policy shock and indicate how rapidly and persistently credit markets incorporate network based exposure into borrowing costs.

The estimated coefficients are plotted against event time and normalized to zero in

Figure 3: Event Study Estimates Around the Paris Agreement – Network Exposure



Notes: This figure plots event-study coefficients β_s from regressions of firm-level borrowing costs on event-time dummies interacted with a network-brown indicator. Event time is measured relative to 2017, and coefficients are normalized to $t = -1$ (2016). Industries are classified as network-brown if their lagged growth in log network-adjusted emissions lies at or above the 80th percentile of the annual cross-industry distribution. The solid red line reports point estimates; the shaded area shows 90 percent confidence intervals based on firm-clustered standard errors. A vertical dashed line marks the year before the implementation of the Paris Agreement.

$t = -1$. Confidence intervals correspond to 90 percent intervals, constructed as plus or minus 1.64 times the firm clustered standard errors. The vertical reference line marks the implementation year, and the horizontal line at zero corresponds to no differential borrowing cost relative to the baseline period.

The pre treatment coefficients are small in magnitude and statistically indistinguishable from zero, providing no evidence of differential pre trends between network brown and network green firms. This absence of pre trends supports the credibility of the identifying assumption underlying both the DiD and event study specifications.

Following the Paris Agreement, borrowing costs for network brown firms increase relative to the control group. The differential effect strengthens in the first and second years after implementation, with peak increases approaching approximately one percentage point. The estimated effects remain positive over subsequent years, suggesting a persistent repricing of firms embedded in carbon intensive production networks. Although confidence intervals widen at longer horizons, reflecting reduced precision, the overall pattern is consistent with a sustained adjustment rather than a temporary response.

Taken together, the event study evidence reinforces the DiD results. The absence of statistically significant pre treatment differentials combined with a clear post 2017 divergence in borrowing costs supports a causal interpretation. The Paris Agreement triggered a relative increase in borrowing costs for firms positioned within carbon intensive production networks, consistent with the view that transition risk is transmitted through supply chain linkages rather than solely through firms’ direct emissions.

6 Conclusion

This paper develops a framework to understand how climate transition risk affects firms’ financing conditions through production networks. Building on recent advances in the theory of production networks, we construct a general equilibrium model with heterogeneous firms, households, and competitive banks in which environmental shocks generate endogenous default risk and are transmitted through input–output linkages. Because firms rely on intermediate inputs and make production decisions before uncertainty is realized, environmental shocks affect not only directly exposed firms but also their suppliers and customers. Competitive banks price these risks through a zero-profit condition, implying that borrowing costs emerge endogenously as equilibrium outcomes reflecting both direct and inherited exposure to environmental risk.

The model delivers a simple prediction. Firms’ borrowing costs depend not only on their own environmental exposure, but also on the environmental characteristics of the production networks in which they operate. As a result, even firms with relatively low direct emissions may face higher borrowing costs when they are connected to carbon-intensive upstream sectors. Transition risk therefore becomes a network phenomenon: environmental shocks propagate through supply chains and are amplified through equilibrium financial conditions.

We bring these predictions to the data by combining firm-level borrowing costs and balance-sheet information with sector-level emissions and detailed U.S. input–output linkages. This allows us to construct a model-consistent measure of network-adjusted carbon exposure that captures firms’ inherited exposure to transition risk through their suppliers.

We first calibrate the model to data and show that it matches the spread between the borrowing costs of green and brown sectors in the US manufacturing industry. The ability of the model to match the spread relies on both the original exposure to shocks and the greater interconnectedness among green sectors on the one side, brown on the other, without perfect separation.

Our regression results show that both direct emissions and network-adjusted emissions are positively associated with borrowing costs. More importantly, the network component remains economically and statistically important even after controlling for firms' own emissions, indicating that lenders account for firms' positions within production networks when pricing credit risk.

Taken together, the difference-in-differences and event-study analyses provide consistent causal evidence that the Paris Agreement led to a repricing of corporate credit driven primarily by firms' exposure to carbon-intensive production networks. While firms with higher direct emissions appear to face higher borrowing costs when considered in isolation, this effect is not robust once network exposure is taken into account. Instead, both empirical strategies indicate that lenders respond most strongly to the carbon intensity embedded in firms' upstream supply chains, reflecting the transmission of transition risk through production networks rather than through firms' own emissions alone.

The sharp timing of the response around the entry into force of the Paris Agreement, combined with the absence of differential pre-trends, supports a causal interpretation in which climate policy expectations are rapidly internalized by credit markets. Borrowing costs increase disproportionately for firms positioned downstream of carbon-intensive suppliers, even when their own direct emissions remain low. These findings underscore that climate transition risk operates as a systemic and forward-looking phenomenon and that credit markets price not only firm-level environmental footprints, but also inherited exposure to upstream carbon risk along the supply chain.

The framework developed here provides a foundation for studying how environmental shocks interact with production networks and financial markets, and how these interactions shape the allocation of credit and the transmission of risk across firms and sectors, in the presence of climate risk. The model could offer clear policy hints, in order to speed up the transition to a low-carbon economy. The fact that the interest rate to green firms is smaller than the one for brown ones increases expected profits and lowers default probability of the green ones. One could make the trade off between these benefits, together with greater TFP, against the cost of the transition from brown to green, explicit, and study how to speed up the transition.

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Appendix

A Model

This appendix provides additional material related to the theoretical framework developed in Section 2. We first collect the proofs of all theoretical results presented in the main text. We then examine whether the key assumptions underlying the model are supported by the empirical production network used in the quantitative and empirical analyses.

In particular, we verify the network and labor-share conditions required by Theorem 1 and Theorem 2 using U.S. input-output data. This exercise serves two purposes. First, it clarifies the economic intuition behind the theoretical results. Second, it provides evidence that the mechanisms generating differences in borrowing costs between green and brown sectors are empirically relevant in the data.

A.1 Proofs

Proof of theorem 1

Proof. Recall that primitive shocks η to green firms dominate those to brown firms in first-order stochastic dominance. For brevity, write $\eta_g \geq \eta_b$. This implies:

$$P(\eta_g \leq x) \leq P(\eta_b \leq x) \quad \text{for all } x \quad (58)$$

where P is the probability on Ω . Since $\rho = L\eta$, the same dominance holds for total (network-propagated) shocks:

$$P(\rho_g \leq Lx) \leq P(\rho_b \leq Lx). \quad (59)$$

Applying monotonic transformations:

$$P(\exp(\rho_g) \leq \exp(Lx)) \leq P(\exp(\rho_b) \leq \exp(Lx)). \quad (60)$$

Dividing by $\mathbb{E}[\exp(\rho_b)]$ preserves the inequality:

$$P\left(\frac{\exp(\rho_g)}{\mathbb{E}[\exp(\rho_b)]} \leq \frac{\exp(Lx)}{\mathbb{E}[\exp(\rho_b)]}\right) \leq P\left(\frac{\exp(\rho_b)}{\mathbb{E}[\exp(\rho_b)]} \leq \frac{\exp(Lx)}{\mathbb{E}[\exp(\rho_b)]}\right). \quad (61)$$

Using the fact that first-order stochastic dominance implies $\mathbb{E}[u(\rho_g)] \geq \mathbb{E}[u(\rho_b)]$ for all non-decreasing functions u , and taking $u = \exp$, we also have:

$$P\left(\frac{\exp(\rho_g)}{\mathbb{E}[\exp(\rho_g)]} \leq \frac{\exp(Lx)}{\mathbb{E}[\exp(\rho_b)]}\right) \leq P\left(\frac{\exp(\rho_g)}{\mathbb{E}[\exp(\rho_b)]} \leq \frac{\exp(Lx)}{\mathbb{E}[\exp(\rho_b)]}\right) \quad (62)$$

and therefore

$$P\left(\frac{\exp(\rho_g)}{\mathbb{E}[\exp(\rho_g)]} \leq \frac{\exp(Lx)}{\mathbb{E}[\exp(\rho_b)]}\right) \leq P\left(\frac{\exp(\rho_b)}{\mathbb{E}[\exp(\rho_b)]} \leq \frac{\exp(Lx)}{\mathbb{E}[\exp(\rho_b)]}\right). \quad (63)$$

for all values of x and therefore all non negative real values that obtain from $\frac{\exp(Lx)}{\mathbb{E}[\exp(\rho_b)]}$. This shows that normalized shocks to green firms dominate those to brown firms:

$$\tau_g \geq \tau_b. \quad (64)$$

If the conditions in the theorem are true, then:

$$\epsilon_g \geq \epsilon_b. \quad (65)$$

Since the min is weakly increasing in both arguments, it follows that:

$$\mathbb{E}[\min(\tau_g, \epsilon_g)] \geq \mathbb{E}[\min(\tau_b, \epsilon_b)], \quad (66)$$

which implies:

$$r_g = \frac{1}{\mathbb{E}[\min(\tau_g, \epsilon_g)]} - 1 \leq \frac{1}{\mathbb{E}[\min(\tau_b, \epsilon_b)]} - 1 = r_b. \quad (67)$$

for any brown firm $b > n_1$ and green firm $g \leq n_1$. ■

Proof of Theorem 2

Proof. We established in the previous theorem that the first order stochastic dominance of the primitive shocks is preserved by the total shocks

$$\tau_g \geq \tau_b. \quad (68)$$

Call M the minimum of the support of the total shocks to green sectors, and m the same

quantity for brown sectors,

$$M = \min_{j \leq n_1} \text{support}(\tau_j) \quad (69)$$

$$m = \min_{j > n_1} \text{support}(\tau_j) \quad (70)$$

and observe that

$$\sum_{j \leq n_1} A_{gj} \tau_j \geq M \sum_{j \leq n_1} A_{gj} \quad (71)$$

$$\sum_{j > n_1} A_{gj} \tau_j \geq m \sum_{j > n_1} A_{gj} \quad (72)$$

$$m \leq M \quad (73)$$

$$\epsilon_g \geq M \sum_{j \leq n_1} A_{gj} + m \sum_{j > n_1} A_{gj} + \beta_g \quad (74)$$

$$\epsilon_b \geq M \sum_{j \leq n_1} A_{bj} + m \sum_{j > n_1} A_{bj} + \beta_b \quad (75)$$

If $\sum_{j \leq n_1} A_{gj} \geq \sum_{j \leq n_1} A_{bj}$, then $\sum_{j > n_1} A_{gj} + \beta_g \leq \sum_{j > n_1} A_{bj} + \beta_b$, because of (4); since $M \geq m$, then the lower bound for ϵ_g is greater than the one for ϵ_b and therefore $\min(\epsilon_g, \tau_g) \geq \min(\epsilon_b, \tau_b)$. The expectation follows the same inequality, and therefore $r_g \leq r_b$. ■

Proof of Corollary 1

Proof. Under the assumptions of the lemma, $\sum_{j \leq n_1} A_{bj} = 0$ and $\sum_{j > n_1} A_{gj} = 0$ for all indices $g \leq n_1, b > n_1$. The conclusion is straightforward. ■

Proof of Corollary 2

Proof. Under the stated assumptions, $\tau_g = 1, \epsilon_g = 1$. The conclusion is straightforward. ■

A.2 Empirical Verification of Theoretical Conditions

This section empirically evaluates the key sufficient conditions underlying Theorem 1 and Theorem 2 of the model. The objective is to assess whether the structural assumptions implied by the theoretical framework are supported by the data once sectors are classified according to their network-adjusted carbon exposure.

The analysis focuses on two dimensions. First, we evaluate whether green sectors are more labor intensive than brown sectors, as required by the theoretical model. Second, we investigate whether production networks display systematic patterns of linkage across environmentally similar sectors. In particular, we test whether green sectors are more strongly connected to other green sectors through input-output relationships. ¹³

A.2.1 Empirical Verification of the Theorems' Conditions

This subsection describes how we empirically verify whether the conditions underlying Theorem 1 and Theorem 2 are supported by the calibrated manufacturing production network.

We begin by classifying sectors as green or brown using the median of **Network CO2** in each year t . Sector k is classified as green if

$$\text{Network CO2}_{k,t} < \text{median}(\text{Network CO2}_t),$$

and brown otherwise. This procedure generates year-specific sets of green and brown sectors, denoted respectively by g_t and b_t .

For each year t , we then use the calibrated manufacturing production network matrix

$$A_t = \begin{cases} A_{2012}^M & \text{if } t \leq 2014, \\ A_{2017}^M & \text{if } t \geq 2015. \end{cases}$$

The matrix is oriented so that $A_{kj,t}$ measures the importance of input j in the production of sector k . Rows therefore correspond to producing sectors, while columns correspond to input suppliers.

We first verify the labor-share condition common to both Theorem 1 and Theorem 2. For each sector k , the implied labor share is computed as

$$\beta_{k,t} = 1 - \sum_j A_{kj,t}.$$

We then compute the average labor share among green and brown sectors:

$$\bar{\beta}_{g,t} = \frac{1}{|G_t|} \sum_{g \in G_t} \beta_{g,t}, \quad \bar{\beta}_{b,t} = \frac{1}{|B_t|} \sum_{b \in B_t} \beta_{b,t}.$$

¹³We use the same data construction we use for the empirics, so refer to Appendix C.1.

The empirical labor-share condition is considered satisfied whenever

$$\bar{\beta}_{g,t} \geq \bar{\beta}_{b,t}.$$

We next evaluate the network condition underlying Theorem 1. Theorem 1 corresponds to a uniform-network benchmark in which green sectors allocate equal importance to green and brown suppliers. Let $\mathcal{G}$ denote the set of Green sectors and $\mathcal{B}$ the set of Brown sectors. The classification is based on the median of network-adjusted emissions, consistently with the empirical analysis in the paper. For each green sector $k \in \mathcal{G}_t$, we compare the total importance of green inputs with the total importance of brown inputs:

$$\sum_{j \in \mathcal{G}} A_{kj,t} = \sum_{j \in \mathcal{B}} A_{kj,t}.$$

Because the empirical matrices are numerical objects, equality is evaluated up to a tolerance level $\varepsilon = 10^{-8}$. For each green sector k , the code therefore defines the indicator

$$I_{k,t}^{(1)} = \mathbf{1} \left\{ \left| \sum_{j \in \mathcal{G}} A_{kj,t} - \sum_{j \in \mathcal{B}} A_{kj,t} \right| \leq \varepsilon \right\}.$$

The year-level Theorem 1 statistic is then computed as

$$S_t^{(1)} = \frac{1}{|\mathcal{G}|} \sum_{k \in \mathcal{G}} I_{k,t}^{(1)}.$$

The equality condition of Theorem 1 is considered satisfied only if

$$S_t^{(1)} = 1,$$

meaning that the condition holds for every green sector in the sample.

The empirical analogue of Theorem 1 is therefore recorded as holding in year t whenever both

$$\bar{\beta}_{g,t} \geq \bar{\beta}_{b,t} \quad \text{and} \quad S_t^{(1)} = 1$$

are simultaneously satisfied.

We then evaluate the network condition underlying Theorem 2. Theorem 2 requires green sectors to be more strongly connected to green suppliers than brown sectors are. To operationalize this condition empirically, we proceed in two steps.

First, for each green sector $g \in \mathcal{G}$, we compute the total importance of green suppliers in the production of sector g :

$$\sum_{j \in \mathcal{G}} A_{gj,t}.$$

We then compare this quantity pairwise with the corresponding quantity for every brown sector $b \in B_t$:

$$\sum_{j \in \mathcal{G}} A_{gj,t} \geq \sum_{j \in \mathcal{G}} A_{bj,t}.$$

This comparison evaluates whether a given green sector relies more heavily on green suppliers than a brown sector does.

For each green sector g , the code computes the share of brown-sector comparisons for which this inequality is satisfied:

$$s_{g,t}^{(2)} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \mathbf{1} \left\{ \sum_{j \in \mathcal{G}} A_{gj,t} \geq \sum_{j \in \mathcal{G}} A_{bj,t} \right\}.$$

Thus, $s_{g,t}^{(2)}$ measures the fraction of brown sectors for which the green sector g exhibits a larger dependence on green suppliers.

Second, the code applies a majority-rule criterion. A green sector g is considered to satisfy the Theorem 2 network condition if

$$s_{g,t}^{(2)} > 0.50.$$

That is, the green sector must dominate more than half of all brown sectors in pairwise comparisons of dependence on green suppliers.

The year-level statistic is then constructed as the share of green sectors satisfying this majority-rule condition:

$$S_t^{(2)} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \mathbf{1} \left\{ s_{g,t}^{(2)} > 0.50 \right\}.$$

The empirical analogue of the Theorem 2 network condition is considered satisfied only if

$$S_t^{(2)} = 1,$$

meaning that every green sector satisfies the pairwise majority-rule criterion.

Finally, the empirical analogue of Theorem 2 is recorded as holding whenever both

$$\bar{\beta}_{g,t} \geq \bar{\beta}_{b,t} \quad \text{and} \quad S_t^{(2)} = 1$$

are satisfied.

The script stores the year-level statistics, including the number of green and brown sectors, average labor shares, the Theorem 1 equality statistic $S_t^{(1)}$, the Theorem 2 pairwise-majority statistic $S_t^{(2)}$, and the final indicators for whether the empirical analogues of Theorem 1 and Theorem 2 hold. These quantities are then averaged across years and reported in the summary table.

A.2.2 Results

To evaluate the empirical relevance of the sufficient conditions derived in Theorem 1 and Theorem 2, we directly test the corresponding restrictions using the empirical input-output matrices combined with the classification of sectors into green and brown groups based on network-adjusted emissions.

Table 8 reports the results.

Labor Share Condition. The labor-share condition receives full empirical support. Across all years in the sample, green sectors exhibit weakly higher average labor shares than brown sectors, implying that the condition $\beta_g \geq \beta_b$ holds in 100% of cases.

This finding is economically meaningful because it suggests that greener sectors rely relatively less on intermediate-input-intensive production structures and more heavily on labor relative to brown sectors.

Theorem 1: Uniform Network Structure. The empirical evidence strongly rejects the network condition underlying Theorem 1. The fraction of green sectors satisfying the equality condition

$$\sum_{j \in \mathcal{G}} A_{kj} = \sum_{j \in \mathcal{B}} A_{kj}$$

is effectively zero in the data.

This result implies that green sectors do not allocate equal input shares across green and brown suppliers, contradicting the symmetric network structure assumed in Theorem 1. Consequently, despite the fact that the labor-share condition holds empirically, the overall sufficient conditions required by Theorem 1 are not supported in the data.

Table 8: Empirical Verification of Theorems' Conditions

	Share satisfying condition
Beta condition	
Avg. $\beta_g \geq \beta_b$	100.0%
Theorem 1	
Equality condition: $\sum_{j \in G} A_{kj} = \sum_{j \in B} A_{kj}$, $k \in G$	0.0%
Theorem 1 holds	No
Theorem 2	
Share satisfying pairwise dominance condition: $\sum_{j \in G} A_{gj} \geq \sum_{j \in G} A_{bj}$, $g \in G$, $b \in B$	86.1%
Theorem 2 holds	Yes

Notes: This table evaluates the empirical validity of the sufficient conditions underlying Theorem 1 and Theorem 2 using the empirical U.S. production network. Sectors are classified as green or brown according to the median level of network-adjusted CO₂ emissions at the six-digit NAICS level. Labor shares are recovered from the production network as $\beta_k = 1 - \sum_j A_{kj}$, where A_{kj} denotes the empirical input-share matrix. The first panel evaluates the labor-share condition common to both theorems. The second panel tests the network symmetry condition underlying Theorem 1, namely whether green sectors allocate equal input shares to green and brown suppliers. The third panel tests the stronger within-group linkage condition underlying Theorem 2 by measuring whether green sectors rely relatively more on inputs originating from other green sectors than from brown sectors. Reported statistics correspond to the fraction of sectors satisfying each condition, averaged across years.

More broadly, the evidence indicates that real-world production networks are highly heterogeneous and far from the uniform benchmark embedded in Theorem 1.

Theorem 2: Stronger Within-Group Green Linkages. In contrast, the network condition underlying Theorem 2 receives substantial empirical support. Approximately 86.1% of pairwise comparisons satisfy the dominance condition requiring that green sectors rely more heavily on green suppliers than comparable brown sectors.

As a result, the sufficient conditions associated with Theorem 2 are empirically supported overall.

These findings indicate the presence of strong within-group clustering in the production network: green sectors are disproportionately connected to other green sectors through upstream input relationships. This structure substantially limits the transmission of transition risk from brown sectors toward green sectors.

B Quantitative Analysis and Calibration

This appendix explains step by step how the primitive downside-shock parameters λ_G and λ_B are calibrated from observed sectoral borrowing costs. The goal of the calibration is to choose the primitive shock intensities so that the equilibrium borrowing costs generated by the model match the average borrowing costs observed in the data for Green and Brown sectors.

Recall from Section 2 that borrowing costs are not imposed exogenously. Instead, they arise endogenously from primitive downside shocks, their propagation through the production network, and the bank zero-profit condition. The calibration therefore maps the observed Green-Brown borrowing-cost spread into differences in primitive downside risk.

Step 1: Construct sector-level observed borrowing costs. For each NAICS6 sector i , we observe borrowing costs $r_{i,t}^{\text{obs}}$. We first aggregate the data at the sector level by computing the within-sector average across time:

$$r_i^{\text{obs}} = \frac{1}{T_i} \sum_{t=1}^{T_i} r_{i,t}^{\text{obs}}. \quad (76)$$

Step 2: Classify sectors as Green or Brown. Sectors are classified using network-adjusted emissions. Let $\mathcal{G}$ denote the set of Green sectors and $\mathcal{B}$ the set of Brown sectors. The classification is based on the median of network-adjusted emissions, consistently with the empirical analysis in the paper.

Using this classification, we compute the average observed borrowing cost within each group:

$$\bar{r}_g^{\text{obs}} = \frac{1}{|\mathcal{G}|} \sum_{i \in \mathcal{G}} r_i^{\text{obs}}, \quad (77)$$

$$\bar{r}_b^{\text{obs}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} r_i^{\text{obs}}. \quad (78)$$

These two empirical moments constitute the calibration targets.

Step 3: Impose a parsimonious two-parameter structure. Rather than estimating a different primitive shock distribution for each sector, the calibration imposes a parsimonious two-parameter structure. All Green sectors share the same primitive downside-shock

intensity λ_G , while all Brown sectors share the same primitive downside-shock intensity λ_B :

$$\lambda_i = \begin{cases} \lambda_g, & i \in \mathcal{G}, \\ \lambda_b, & i \in \mathcal{B}. \end{cases} \quad (79)$$

The entire calibration therefore reduces to estimating only two parameters.

Step 4: Draw primitive downside shocks. Recall from the theoretical model that primitive shocks follow exponential downside distributions. For each candidate parameter pair (λ_g, λ_b) , the simulation draws

$$X_i^{(m)} \sim \text{Exp}(\lambda_i), \quad m = 1, \dots, M, \quad (80)$$

and defines the primitive log-productivity shock as

$$\eta_i^{(m)} = -X_i^{(m)} \leq 0. \quad (81)$$

The parameter λ_i controls the severity of downside risk because

$$\mathbb{E}[X_i] = \frac{1}{\lambda_i}. \quad (82)$$

A lower value of λ_i therefore corresponds to larger expected downside shocks. Consequently, if brown sectors are calibrated to be riskier than green sectors, the calibration should imply $\lambda_b < \lambda_g$.

Step 5: Propagate shocks through the production network. Primitive shocks are then propagated through the empirical production network using the Leontief inverse

$$L = (I - A')^{-1}. \quad (83)$$

As shown in the theoretical section, the effective shock affecting sector i is not simply its primitive shock η_i , but the network-adjusted shock

$$\rho_i^{(m)} = \sum_j L_{ij} \eta_j^{(m)}. \quad (84)$$

In vector notation,

$$\rho^{(m)} = L\eta^{(m)}. \quad (85)$$

Thus, a sector is affected not only by its own primitive environmental risk, but also by the shocks transmitted through upstream supplier linkages. This is the key network-propagation mechanism of the model.

Step 6: Construct repayment objects entering the bank problem. Using the propagated shocks, the simulation computes the normalized repayment variable

$$\tau_i^{(m)} = \frac{\exp[\rho_i^{(m)}]}{\frac{1}{M} \sum_{m=1}^M \exp[\rho_i^{(m)}]}. \quad (86)$$

Recall from Section 2 that τ_i captures the normalized repayment capacity of sector i after network propagation.

The simulation then computes the supplier-side repayment component

$$\epsilon_i^{(m)} = \beta_i + \sum_j A_{ji} \tau_j^{(m)}, \quad (87)$$

where β_i is the labor share of sector i .

Economically, ϵ_i summarizes the repayment capacity inherited from upstream suppliers through the production network.

Step 7: Solve the bank zero-profit condition. Banks are competitive and risk neutral. As shown in Section 2, equilibrium borrowing costs are pinned down by the bank zero-profit condition.

Under full debt financing, $\theta_i = 1$, the equilibrium condition can be written as

$$\mathbb{E}[\min(\tau_i, \epsilon_i)] = \frac{1}{1 + r_i^{\text{model}}}. \quad (88)$$

Equivalently,

$$r_i^{\text{model}} = \frac{1}{\mathbb{E}[\min(\tau_i, \epsilon_i)]} - 1. \quad (89)$$

In the Monte Carlo implementation, expectations are approximated numerically:

$$\hat{r}_i^{\text{model}} = \frac{1}{\frac{1}{M} \sum_{m=1}^M \min[\tau_i^{(m)}, \epsilon_i^{(m)}]} - 1. \quad (90)$$

For leverage values $\theta_i \in (0, 1)$, the simulation instead solves numerically for the debt-cost variable ζ_i associated with the nonlinear bank zero-profit condition, and then recovers the simple interest rate using

$$r_i^{\text{model}} = \frac{\exp(\zeta_i) - 1}{\theta_i}. \quad (91)$$

Step 8: Compute model-implied Green and Brown borrowing costs. For each candidate parameter pair (λ_g, λ_b) , the simulation produces sector-level model-implied borrowing costs $r_i^{\text{model}}(\lambda_g, \lambda_b)$.

These sectoral borrowing costs are then averaged separately across green and brown sectors:

$$\bar{r}_g^{\text{model}} = \frac{1}{|\mathcal{G}|} \sum_{i \in \mathcal{G}} r_i^{\text{model}}, \quad (92)$$

and

$$\bar{r}_b^{\text{model}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} r_i^{\text{model}}. \quad (93)$$

These are the model-implied moments matched to the empirical targets.

Step 9: Numerical calibration. The calibrated parameters are chosen to minimize the squared distance between model-implied and observed borrowing costs:

$$(\hat{\lambda}_g, \hat{\lambda}_b) = \arg \min_{\lambda_g > 0, \lambda_b > 0} \mathcal{L}(\lambda_g, \lambda_b), \quad (94)$$

where

$$\begin{aligned} \mathcal{L}(\lambda_g, \lambda_b) = & [\bar{r}_g^{\text{model}} - \bar{r}_g^{\text{obs}}]^2 \\ & + [\bar{r}_b^{\text{model}} - \bar{r}_b^{\text{obs}}]^2. \end{aligned} \quad (95)$$

To guarantee positivity of the parameters, the optimization is performed in logarithmic space. The numerical algorithm therefore optimizes over

$$a_g = \log \lambda_g, \quad a_b = \log \lambda_b, \quad (96)$$

and then recovers the economically meaningful parameters through exponentiation:

$$\lambda_g = \exp(a_g), \quad \lambda_b = \exp(a_b). \quad (97)$$

The implementation uses a Nelder-Mead simplex algorithm combined with Monte Carlo simulation of repayment outcomes.

C Data

This appendix provides a detailed description of the datasets, variable construction procedures, and data-processing steps used throughout the paper. The objective is to construct measures of environmental exposure that are consistent with the theoretical framework developed in Section 2 and can be employed both in the quantitative calibration and in the empirical analysis.

The sections that follow describe the individual data sources, the construction of direct and network-adjusted emissions, and the definition of the financial variables used in the empirical analysis.

C.1 CO₂ Emissions Data

This section of the appendix provides a detailed and rigorous description of the CO₂ emissions data sources, transformations, and calibration procedures used in the paper. The objective is to map the theoretical objects defined in the model to empirically observable counterparts in a manner that is transparent, internally consistent, and replicable. We describe in turn the construction of direct and network-adjusted emissions and the calibration of the production network matrix used in the model.

Throughout the appendix, industries are indexed by $k \in \mathcal{V}$, consistently with the notation adopted in the theoretical and empirical sections.

C.1.1 EPA Greenhouse Gas Reporting Program

The primary source of direct emissions data is the U.S. Environmental Protection Agency (EPA) Greenhouse Gas Reporting Program (GHGRP). The GHGRP was established in 2008 under the authority of the Clean Air Act with the objective of providing comprehensive, facility-level information on greenhouse gas emissions across the U.S. economy. The program is designed to support federal and state climate policy, improve emissions transparency, and provide a consistent statistical foundation for the analysis of greenhouse gas emissions across industries and over time.

Under the GHGRP, reporting is mandatory for facilities operating in the United States whose annual direct greenhouse gas emissions exceed 25,000 metric tons of carbon dioxide equivalent (CO₂e). In addition to large stationary sources, the program also covers selected upstream suppliers of fossil fuels and industrial gases whose products give rise to downstream

emissions. As a result, the GHGRP captures the vast majority of emissions generated by energy-intensive activities, including manufacturing, electricity generation, petroleum refining, chemical production, metals, minerals, and other heavy industrial sectors. While smaller facilities fall below the reporting threshold, the GHGRP is widely recognized as covering a dominant share of industrial greenhouse gas emissions in the United States.

Emissions are reported annually at the facility level and encompass all major greenhouse gases regulated under the program, including carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), and fluorinated gases. Reported quantities of individual gases are converted into a common metric of carbon dioxide equivalent using 100-year global warming potentials consistent with the Intergovernmental Panel on Climate Change Fourth Assessment Report. This conversion allows emissions from different gases to be aggregated into a single, economically meaningful measure of climate impact. Emissions are expressed in metric tons of CO_2 equivalent and reflect direct emissions generated by on-site fuel combustion, industrial processes, and other covered activities.

A key feature of the GHGRP is the detailed industry classification reported for each facility. Each reporting entity is assigned a Primary NAICS Code, corresponding to the six-digit North American Industry Classification System. This classification reflects the facility's main economic activity and provides a natural bridge between facility-level emissions data and industry-level economic data. The availability of consistent NAICS identifiers makes it possible to aggregate emissions across facilities to construct sector-level emissions measures that align with standard input–output and national accounts data.

In this paper, we use the EPA-provided multi-year summary file `ghgp_data_by_year_2023.xlsx`, which consolidates facility-level GHGRP data for reporting years 2010 through 2023. Within this file, we rely on the worksheet labeled Direct Point Emitters, which contains annual total direct emissions for each reporting facility along with its Primary NAICS Code and other identifying information. For each facility and year, the dataset reports the total quantity of direct greenhouse gas emissions, expressed in metric tons of CO_2 equivalent.

The GHGRP is particularly well suited for the analysis conducted in this paper for several reasons. First, it provides emissions data at a high level of disaggregation, allowing emissions to be mapped precisely to six-digit NAICS industries. Second, the use of a consistent reporting threshold and standardized measurement methodology ensures comparability across facilities and over time. Third, the focus on direct emissions aligns closely with the concept of sector-specific productivity shocks in the model, which operate through industries' own

production activities. Finally, the annual frequency of the data allows emissions measures to be matched directly to the timing of financial variables and policy events considered in the empirical analysis.

Aggregation to Six-Digit NAICS Industries Facility-level emissions are aggregated to the six-digit NAICS industry level in order to align emissions data with the production network used in the model. This choice is primarily dictated by the structure of the BEA input-output tables, which represent inter-industry linkages at the level of detailed industries that can be consistently mapped to six-digit NAICS classifications. As a result, the six-digit NAICS level constitutes the most granular level at which direct emissions from the EPA GHGRP can be merged with the BEA production network in a consistent and economically meaningful manner.

Let $e_{f,t}$ denote the total direct greenhouse gas emissions of facility f in year t , measured in metric tons of CO₂ equivalent. Each facility is assigned to an industry $k(f)$ based on its Primary NAICS Code reported in the GHGRP. Aggregating emissions across facilities within the same six-digit NAICS industry yields a sector-level measure of direct emissions.

Formally, sector-level direct emissions are constructed as

$$\text{CO2}_{k,t} = \sum_{f \in k} e_{f,t}, \quad (98)$$

where the summation runs over all facilities whose primary NAICS code corresponds to sector k . This aggregation preserves the full coverage of facility-level emissions while producing industry-level measures that are directly comparable across sectors and over time.

The aggregation is implemented in the Python script `0_CO2_Extraction.py`. The script programmatically identifies the header row of the EPA spreadsheet, extracts all year-specific emissions columns, reshapes the data to long format, and aggregates emissions across facilities within each six-digit NAICS sector and year. The resulting panel dataset contains one observation per industry k and year t , with $\text{CO2}_{k,t}$ expressed in metric tons of CO₂e.

Aggregating emissions at the six-digit NAICS level ensures consistency with the BEA input-output tables used to construct the production network and allows emissions to be propagated through the economy using the Leontief framework described in the subsequent subsection. In the empirical analysis, emissions are rescaled and reported in million metric tons of CO₂e for ease of interpretation, without affecting the underlying construction.

C.1.2 BEA Input-Output Tables

To capture production network linkages, we rely on industry-by-industry input–output tables produced by the U.S. Bureau of Economic Analysis. Specifically, we use the total requirements matrices provided in the file `IxI_TR_Detail.xlsx`, which contains benchmark-year tables for 2007, 2012, and 2017.

For a given benchmark year y , the BEA total requirements matrix L_y is defined such that each element $L_{jk,y}$ measures the total dollar value of output from industry j required, directly and indirectly, to deliver one dollar of final demand for the output of industry k . These matrices therefore capture the full upstream propagation of production through the economy.

The Python script `0_IO_Tables_Extraction.py` extracts the numerical blocks corresponding to the total requirements matrices and the associated industry codes. Rows and columns corresponding to subtotals or non-industry entries are removed. The resulting matrices are square and indexed by industry codes that can be mapped to six-digit NAICS sectors.

C.1.3 From BEA Total Requirements to Model Production Networks

This subsection explains in detail how the production network matrix used in the model is constructed from the BEA industry-by-industry total requirements tables and clarifies the distinction between empirical input–output objects and their model-consistent counterparts.

The BEA industry-by-industry total requirements table for a given benchmark year y is a square matrix, denoted $\mathbf{L}_y^{TR}$. In the Excel file `IxI_TR_Detail.xlsx`, each table appears as a rectangular numerical block in which rows correspond to supplying industries and columns correspond to industries delivering final demand. The Python script `0_IO_Tables_Extraction.py` reads the industry labels from the first row and first column of this block and constructs a square matrix indexed consistently across rows and columns. An entry $L_{jk,y}^{TR}$ therefore corresponds to industry j as a supplier (row) and industry k as a demander (column).

Each column k of $\mathbf{L}_y^{TR}$ reports the total gross output required from all industries in order to deliver one dollar of final demand for the output of industry k . In particular, the element $L_{jk,y}^{TR}$ measures the total dollar value of output from industry j that is required, both directly and indirectly through all upstream production stages, to produce one dollar of final demand for industry k . This column-wise interpretation is crucial for mapping the BEA object into the production network structure of the model.

In the model, the importance coefficient A_{jk} captures the share of sector j 's output used as an intermediate input in the production of sector k . Columns of the matrix $\mathbf{A}$ therefore describe the input composition of sector k , while rows describe how the output of a given sector j is distributed across downstream users. Under the standard Leontief production framework, the relationship between the BEA total requirements matrix and the matrix of empirical direct input coefficients can be written as

$$\mathbf{L}_y^{TR} = (\mathbf{I} - \mathbf{A}_y^{TR})^{-1}, \quad (99)$$

where $\mathbf{A}_y^{TR}$ denotes the matrix of direct requirements implied by the BEA data and $\mathbf{I}$ denotes the identity matrix. This relationship reflects the fact that total requirements incorporate not only direct input needs but also all higher-order indirect requirements arising from upstream production linkages.

Given the BEA total requirements matrix, the implied empirical direct requirements matrix is therefore recovered as

$$\mathbf{A}_y^{TR} = \mathbf{I} - (\mathbf{L}_y^{TR})^{-1}. \quad (100)$$

This inversion step maps the observed BEA production network into a matrix of empirical input shares. In the implementation, the script `0_I0_C02_Merge.py` loads the cleaned numerical block for $\mathbf{L}_y^{TR}$, verifies that row and column labels coincide, computes its numerical inverse, and constructs $\mathbf{A}_y^{TR}$ accordingly.

The matrix $\mathbf{A}_y^{TR}$ does not yet correspond to the production network used in the model, as the latter incorporates financial frictions that distort observed input shares. Following the calibration procedure described in Appendix B, we obtain the model-consistent production network matrix $\mathbf{A}_y$ by rescaling $\mathbf{A}_y^{TR}$ using sector-specific interest rate derived from observed borrowing costs.

A key theoretical restriction in the model, stemming from the constant returns-to-scale normalization of industries' production functions, is that the total value of intermediate inputs used by any industry cannot exceed the value of its gross output. In matrix terms, this restriction implies that no column sum of the matrix $\mathbf{A}_y$ exceeds one. As a consequence, $\mathbf{A}_y$ is a column-substochastic matrix, and its spectral radius, defined as the largest modulus of its eigenvalues, is weakly less than one. Throughout the analysis, we assume that the spectral radius of $\mathbf{A}_y$ is strictly less than one. This is a mild and economically natural condition: it is satisfied, for example, if every industry is connected, possibly through multiple production

stages, to at least one sector that employs a primary factor such as labor.

Under this assumption, the economy's Leontief inverse used in the model is defined as

$$\mathbf{L}_y = (\mathbf{I} - \mathbf{A}'_y)^{-1}, \quad (101)$$

where $\mathbf{A}'_y$ denotes the transpose of $\mathbf{A}_y$. The transpose appears because, in the model, columns of $\mathbf{A}_y$ index users while rows index suppliers, so that the propagation of shocks through upstream linkages naturally involves $\mathbf{A}'_y$. The matrix $\mathbf{L}_y$ can be expressed as the convergent Neumann series

$$\mathbf{L}_y = \sum_{h=0}^{+\infty} (\mathbf{A}'_y)^h. \quad (102)$$

This representation makes explicit that the Leontief inverse aggregates the effects of all production paths of arbitrary length in the network. Each term $(\mathbf{A}'_y)^h$ captures the contribution of paths of length h , and the nonnegativity of $\mathbf{A}_y$ implies that all entries of $\mathbf{L}_y$ are nonnegative. In particular, $L_{kj,y}$ measures the cumulative importance of industry j as an upstream supplier to industry k through all possible chains of intermediate input relationships.

The Python script `0_I0_C02_Merge.py` implements these transformations directly. Starting from the BEA total requirements matrix $\mathbf{L}_y^{TR}$, the script computes $\mathbf{A}_y^{TR}$, rescales it to obtain the model-consistent matrix $\mathbf{A}_y$, and verifies that the resulting matrix satisfies the column-substochasticity and nonnegativity conditions required for the existence of the Leontief inverse. This construction ensures that the empirical production network used in the paper is fully consistent with the theoretical structure of the model and allows productivity and emissions shocks to propagate through the economy according to the Leontief expansion described in Section 2.

C.1.4 Visualization of the Production Network

Figures 4–7 provide graphical representations of the model-consistent production network constructed following the procedure described above. These figures are intended as descriptive illustrations of the structure and intensity of production linkages among manufacturing industries and are not used directly in the quantitative analysis.

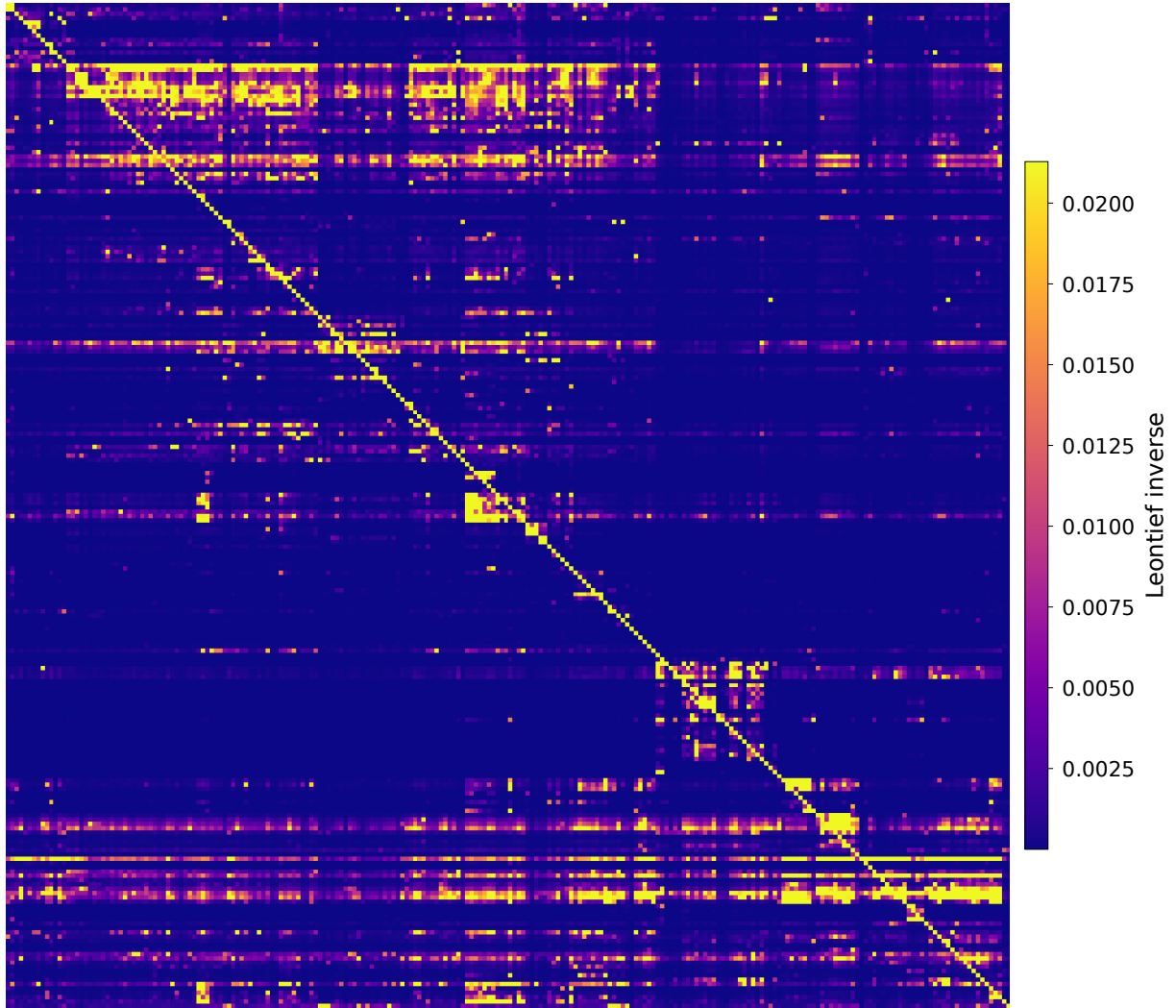
Figures 4 and 5 display heatmap representations of the Leontief inverse matrices $\mathbf{L}_{2012}$ and $\mathbf{L}_{2017}$ for manufacturing industries. Each cell (k, j) is colored according to the magnitude of the element $L_{kj,y}$, which measures the cumulative upstream importance of industry j for

the production of industry k , aggregating direct and indirect input requirements along all production paths in the network. Brighter colors correspond to stronger total requirements. The prominent diagonal reflects the fact that own-industry effects dominate total requirements, while off-diagonal clusters highlight economically meaningful upstream-downstream linkages across manufacturing sectors. Differences in color intensity patterns across years reflect changes in the structure and strength of production interdependencies over time.

Figures 6 and 7 provide network visualizations of the corresponding model-consistent input-output matrices $\mathbf{A}_{2012}$ and $\mathbf{A}_{2017}$. Nodes represent manufacturing industries, and directed edges capture intermediate input linkages from supplying to using industries. Node size and color encode eigenvector centrality, which measures an industry’s importance in the production network by accounting for both the number and the strength of its connections, as well as the centrality of its neighbors. Lighter and larger nodes correspond to industries that are more central in the propagation of shocks through the network. Edge thickness is proportional to the magnitude of the underlying input coefficient, while the absence of labels emphasizes the global structure of the network rather than individual sector identities.

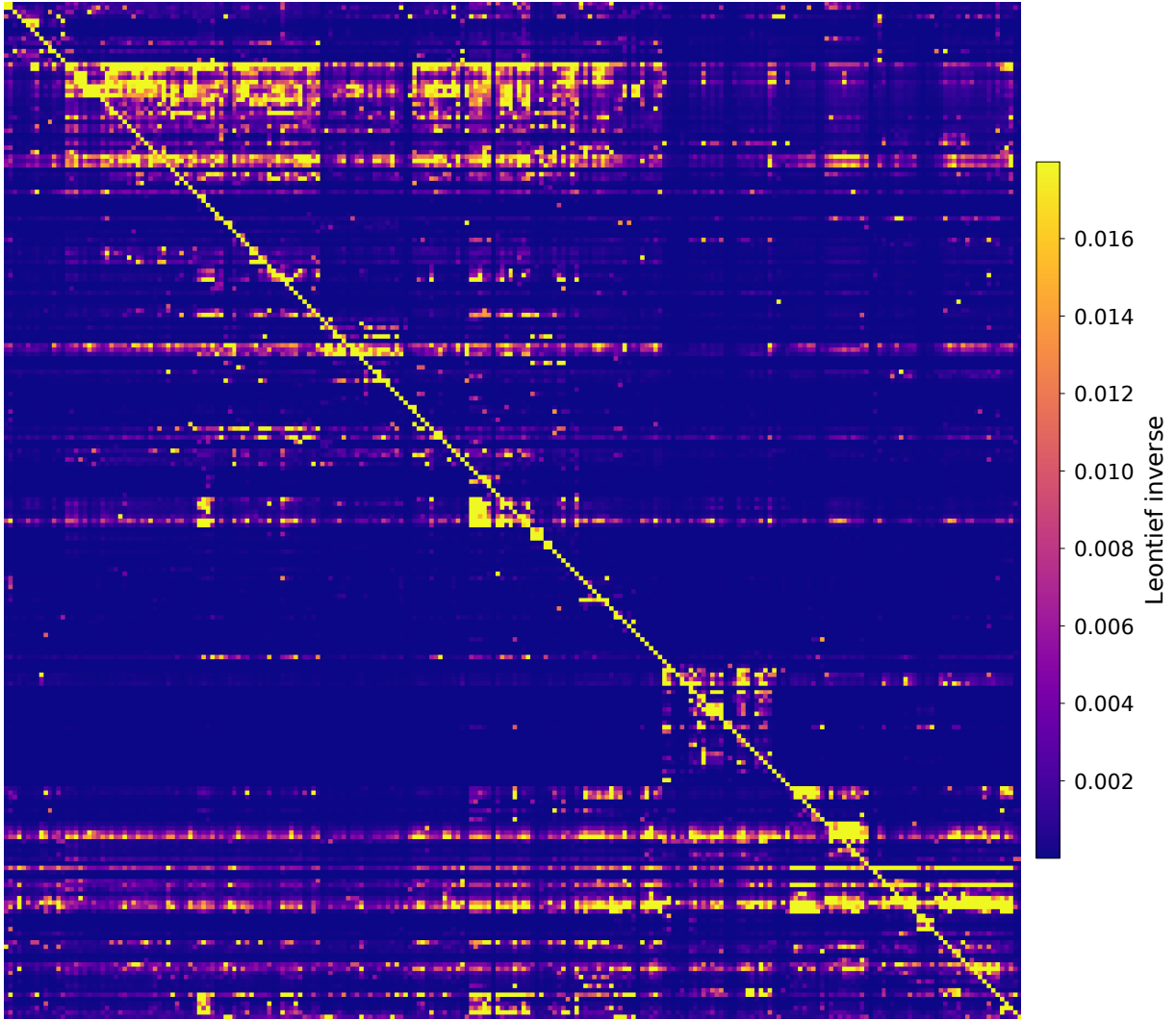
Taken together, these figures illustrate two complementary perspectives on the production network. The Leontief heatmaps emphasize the cumulative strength of upstream dependencies embedded in the economy, while the network diagrams highlight the heterogeneous centrality of industries and the sparsity of economically relevant linkages. Both representations are fully consistent with the theoretical structure of the model and with the construction of the production network matrix described in this appendix.

Figure 4: Leontief Inverse Matrix, Manufacturing Industries, 2012



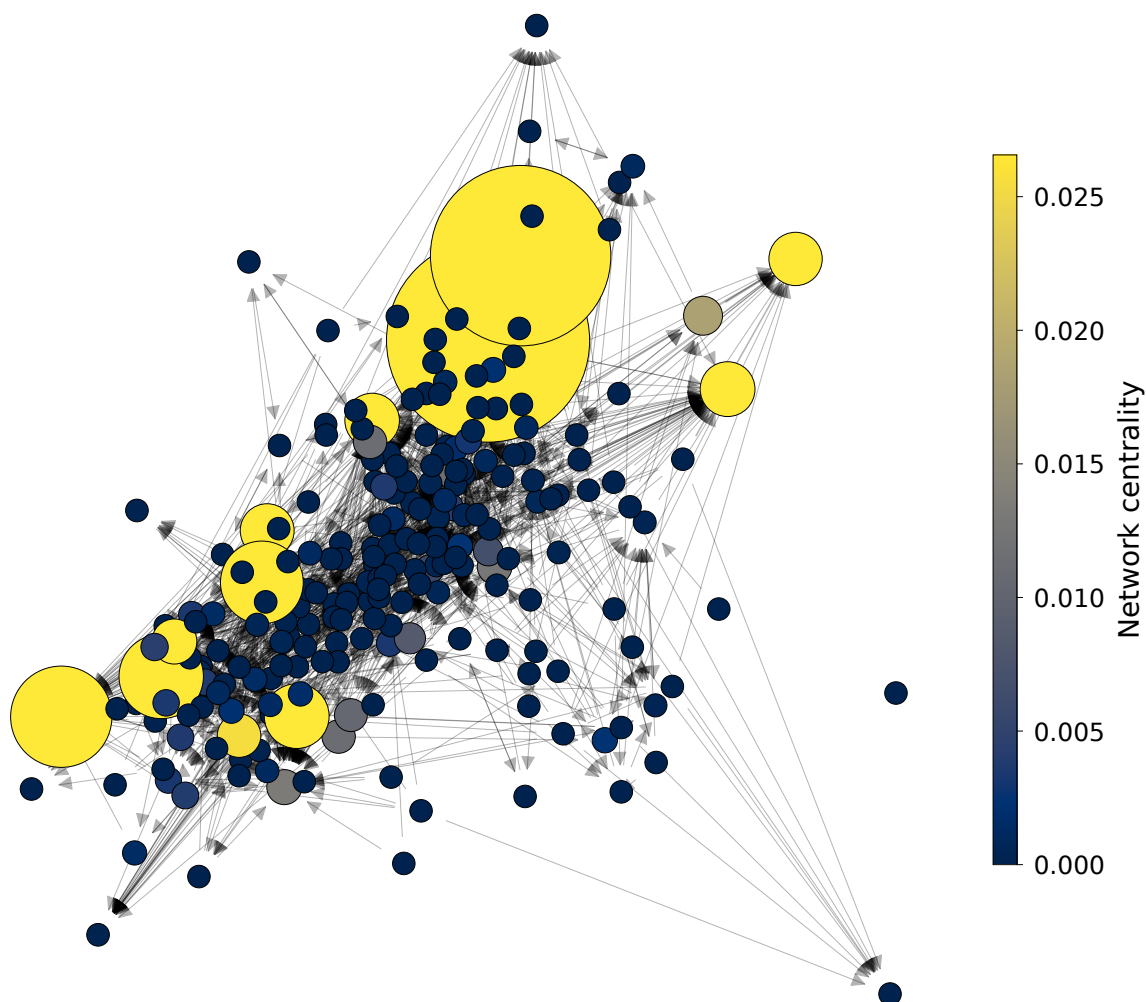
Notes: This figure shows a heatmap of the model-consistent Leontief inverse $\mathbf{L}_{2012} = (\mathbf{I} - \mathbf{A}'_{2012})^{-1}$ restricted to manufacturing industries. Each cell (k, j) is colored according to the magnitude of $L_{kj,2012}$, which measures the cumulative importance of industry j as an upstream supplier to industry k through all direct and indirect input linkages. Brighter colors indicate stronger total (direct and indirect) input requirements.

Figure 5: Leontief Inverse Matrix, Manufacturing Industries, 2017



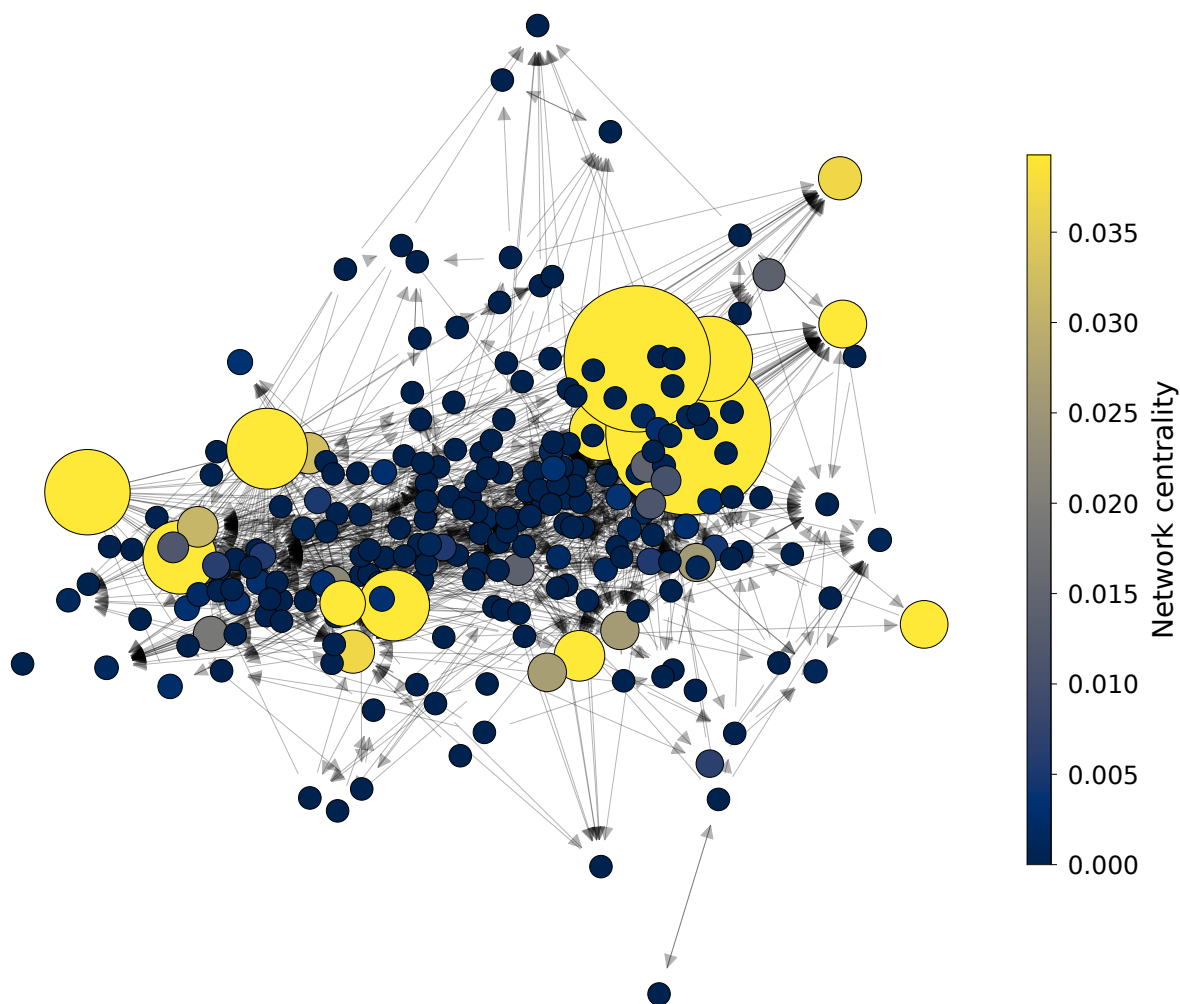
Notes: This figure shows a heatmap of the model-consistent Leontief inverse $\mathbf{L}_{2017} = (\mathbf{I} - \mathbf{A}'_{2017})^{-1}$ restricted to manufacturing industries. Cell colors reflect the magnitude of $L_{kj,2017}$, with brighter cells corresponding to stronger cumulative upstream dependence of industry k on industry j . Differences in the pattern and intensity of off-diagonal elements relative to Figure 4 illustrate changes in the structure of manufacturing production linkages between 2012 and 2017.

Figure 6: Production Network Representation, Manufacturing Industries, 2012



Notes: Nodes represent manufacturing industries and directed edges represent intermediate input linkages in the model-consistent production network $\mathbf{A}_{2012}$. Node size and color encode eigenvector centrality, so that larger and lighter nodes are more central in the network, taking into account both the number and the strength of their connections and the centrality of their neighbors. Edge thickness is proportional to the corresponding input coefficient. The absence of labels emphasizes the overall topology of the production network rather than individual sector identities.

Figure 7: Production Network Representation, Manufacturing Industries, 2017



Notes: This figure provides the network representation of the model-consistent production matrix $\mathbf{A}_{2017}$ for manufacturing industries. As in Figure 6, node size and color reflect eigenvector centrality and edge thickness reflects the strength of intermediate input linkages. Comparing Figures 6 and 7 highlights changes in the centrality of industries and in the density and orientation of production linkages over time.

C.1.5 Calibration of the Production Network and Construction of Network-Adjusted Emissions

This subsection describes in detail how the empirical production network used in the paper is constructed and how it is combined with sector-level emissions data to obtain the network-adjusted emissions measure, denoted by **Network C02**. The objective of this procedure is to construct an empirical counterpart to the theoretical propagation mechanism developed in Section 2, where environmental exposure is transmitted through production linkages and therefore depends not only on a sector’s own emissions, but also on the emissions embodied in its upstream supply chain.

The starting point of the construction is the BEA industry-by-industry total requirements matrix. We use the benchmark systems for 2012 and 2017, denoted respectively by

$$\mathbf{L}_{2012}^{BEA} \quad \text{and} \quad \mathbf{L}_{2017}^{BEA}.$$

These matrices are indexed by NAICS6 industries across both rows and columns. Rows correspond to supplying industries, while columns correspond to using industries. Therefore, each element L_{jk}^{BEA} measures the total amount of output from supplier industry j required, directly and indirectly, to produce one unit of final demand in industry k .

Because the empirical analysis focuses on manufacturing industries only, we cannot directly use the original BEA total requirements matrix. Restricting the original Leontief inverse to manufacturing rows and columns would still retain indirect propagation paths inherited from non-manufacturing sectors embedded in the original inversion. To construct a production network defined exclusively over manufacturing industries, we therefore first recover the underlying direct requirements matrix.

For each benchmark year $s \in \{2012, 2017\}$, we compute

$$\mathbf{A}_s = \mathbf{I} - (\mathbf{L}_s^{BEA})^{-1}.$$

The resulting matrix $\mathbf{A}_s$ contains the direct input coefficients of the economy. Its element A_{jk} measures the share of sector k ’s expenditure devoted to intermediate inputs purchased from supplier sector j . Rows therefore correspond to suppliers, while columns correspond to input users or producing industries.

The orientation of the matrix is important both theoretically and empirically. Since each column corresponds to the production structure of a sector, the columns of $\mathbf{A}_s$ collect all intermediate input shares used in the production of sector k . Labor and value added are

not included among intermediate inputs. As a consequence, the direct requirements matrix must satisfy

$$\sum_j A_{jk} < 1.$$

The residual

$$\beta_k = 1 - \sum_j A_{jk}$$

is interpreted as the labor or value-added share of sector k . In the implementation, we explicitly verify this restriction by computing column sums and implied labor shares for all sectors. Industries generating non-positive residual labor shares are treated as problematic observations and removed from the sample.

The inversion step is also numerically validated. Specifically, after recovering $\mathbf{A}_s$, we compute the condition number of the matrix $(\mathbf{I} - \mathbf{A}_s)$. This diagnostic verifies that the Leontief system is well-conditioned and sufficiently far from singularity, ensuring that the implied production network is numerically stable and economically meaningful.

In input output models a la Acemoglou, without debt, the input output observed matrix represents $\mathbf{A}$ thanks to the first order conditions (FOCs) for firmsâ intermediate goods demand. With leverage, the FOCs are different. We obtain the production network using sector-level borrowing costs. In the theoretical model, the importance coefficients appear modified because of leverage. In particular, the first-order conditions imply

$$\frac{p_j z_{jk}}{p_k y_k} = \frac{A_{jk}}{1 + r_k \theta_k} = A_{jk} \exp(-\zeta_k),$$

where r_k denotes the borrowing rate faced by sector k , θ_k denotes leverage, and $\zeta_k = \log(1 + r_k \theta_k)$. Rearranging yields

$$A_{jk} = \frac{p_j z_{jk}}{p_k y_k} \exp(\zeta_k).$$

Accordingly, we construct a calibrated production network matrix by rescaling the empirical direct requirements matrix using sector-specific borrowing costs. In the implementation, this is done column by column:

$$A_{jk}^{scaled} = A_{jk}(1 + r_k).$$

Intuitively, this adjustment increases the effective importance of inputs used by sectors facing higher borrowing costs and maps the empirical production structure into the model-

consistent network object implied by the presence of financial rigidities.

We then restrict the calibrated production network to manufacturing industries only. Manufacturing industries are defined as NAICS2 sectors beginning with 31, 32, or 33. Let $\mathcal{M}$ denote the set of manufacturing sectors. We construct the manufacturing-only direct requirements matrix by extracting the manufacturing block:

$$\mathbf{A}_s^M = \mathbf{A}_{s,\mathcal{M}\mathcal{M}}^{scaled}.$$

This restriction removes all non-manufacturing industries from both the supplier and user dimensions of the network.

Because restricting the economy to manufacturing changes the structure of the network itself, we then recompute a new manufacturing-only Leontief inverse:

$$\mathbf{L}_s^M = (\mathbf{I} - \mathbf{A}_s^M)^{-1}.$$

This second inversion is essential. The original BEA total requirements matrix measures propagation through the entire economy, including non-manufacturing suppliers. By contrast, the recalculated matrix $\mathbf{L}_s^M$ captures only direct and indirect propagation paths operating within manufacturing supply chains. The resulting object therefore measures manufacturing-to-manufacturing network exposure exclusively.

Once the calibrated manufacturing Leontief inverse has been constructed, we combine it with sector-level emissions data to construct the network-adjusted emissions measure.

Let $\mathbf{CO2}_t$ denote the vector of direct greenhouse gas emissions across manufacturing industries in year t . Each element $\mathbf{CO2}_{k,t}$ corresponds to the total direct emissions of industry k , measured in metric tons of CO₂ equivalent and constructed from the EPA GHGRP as described in Appendix C.1.1. The emissions vector is ordered consistently with the rows and columns of the manufacturing Leontief inverse.

The vector of network-adjusted emissions in year t , denoted by **Network CO2_t**, is then defined as

$$\mathbf{Network\ CO2}_t = \mathbf{L}_t^{M'} \mathbf{CO2}_t.$$

The transpose appears because rows of the production network matrix index suppliers while columns index users. Premultiplication by $\mathbf{L}_t^{M'}$ therefore aggregates upstream exposure from all supplier industries connected to sector k .

Element by element, the network-adjusted emissions of industry k in year t are given by

$$\text{Network CO2}_{k,t} = \sum_{j \in \mathcal{M}} L_{jk,t}^M \text{CO2}_{j,t}.$$

This expression makes explicit that **Network CO2** $_{k,t}$ aggregates the direct emissions of all upstream manufacturing industries j , weighted by their cumulative importance in the production structure of sector k through all direct and indirect production linkages. The manufacturing Leontief inverse therefore accounts not only for direct suppliers, but also for suppliers of suppliers and all higher-order production relationships embedded in the manufacturing network.

In practice, BEA benchmark tables are available only for a limited set of years. The implementation therefore approximates the evolution of the production network over time using the 2012 and 2017 benchmark systems. Specifically,

$$\mathbf{L}_t^M = \begin{cases} \mathbf{L}_{2012}^M & \text{if } t \leq 2014, \\ \mathbf{L}_{2017}^M & \text{if } t \geq 2015. \end{cases}$$

For each year t , the emissions vector is aligned with the row ordering of the selected manufacturing Leontief inverse, and the transformation

$$\text{Network CO2}_t = \mathbf{L}_t^{M'} \text{CO2}_t$$

is applied only to sectors with non-missing emissions observations.

Because the entries of the Leontief inverse are unitless while emissions are measured in metric tons of CO₂ equivalent, the resulting vector **Network CO2** $_t$ is expressed in the same units. In the empirical analysis, both **CO2** and **Network CO2** are sometimes rescaled and reported in million metric tons of CO₂ equivalent for ease of interpretation.

The resulting variable **Network CO2** constitutes the empirical counterpart to the propagation mechanism developed in the theoretical model and allows us to distinguish between direct environmental exposure, captured by **CO2**, and exposure transmitted through production network linkages, captured by **Network CO2**.

C.2 Compustat Firm-Level Financial Data

Firm-level financial variables are obtained from the Compustat Fundamentals Annual database, covering publicly listed U.S. firms. The sample spans the years 2012 to 2023. Firms are included in the analysis only if they report valid six-digit NAICS industry codes and have non-missing entries for interest expense and debt-related variables. Observations with implausible or extreme financial ratios are excluded as described below.

The key financial variables used in our analysis include:

- `xint`: Interest expense
- `dltt`: Long-term debt obligations
- `dlc`: Short-term debt (due within one year)
- `lt`: Total liabilities
- `at`: Total assets
- `che`: Cash and equivalents
- `ebit`: Earnings before interest and taxes
- `ppent`: Net property, plant, and equipment
- `ceq`, `mkvalt`: Variables used to compute market-to-book ratios

To mitigate the influence of outliers, observations with implausible or extreme values of leverage, cash holdings, total assets, and investment rates are trimmed at the top 1 percent of their respective distributions. All financial variables entering the empirical analysis are expressed in logarithms. Observations with missing or non-positive values for key balance-sheet items are excluded. The resulting sample provides a consistent panel of U.S. firms suitable for analyzing firm-level financial conditions using annual Compustat data.

C.2.1 Measuring the Cost of Debt

A central variable in our empirical analysis is the firm-level cost of debt. We construct this measure using firms' realized interest expenses and outstanding debt balances, with the goal of capturing the effective borrowing rate paid on all interest-bearing obligations.

This approach reflects the average cost of debt financing faced by a firm in a given year, aggregating across maturities and debt instruments.

Our preferred measure of the cost of debt is defined as the ratio of interest expense to total interest-bearing debt, expressed as a percentage. Formally, for firm i in year t , we define:

$$r_{i,t} = \frac{\text{xint}_{i,t}}{\text{dltt}_{i,t} + \text{dlc}_{i,t}} \times 100, \quad (103)$$

where $\text{xint}_{i,t}$ denotes interest expense and $\text{dltt}_{i,t} + \text{dlc}_{i,t}$ corresponds to total interest-bearing debt, composed of long-term debt due beyond one year and short-term debt due within one year.

This firm-level measure captures the effective interest rate paid on outstanding debt and is well suited to environments in which borrowing costs may vary systematically across firms due to differences in risk exposure, capital structure, or regulatory conditions. Unlike contractual interest rates on individual instruments, this measure reflects the realized financial burden of debt financing faced by the firm as a whole.

Because the construction of implied interest rates using accounting data can generate extreme values due to reporting errors, temporary debt reductions, or timing mismatches between interest expenses and balance sheet stocks, we implement a multi-step data cleaning procedure. First, we drop observations with missing values for interest expense or interest-bearing debt, as well as firms without valid industry identifiers. Second, we winsorize key balance sheet components, including interest expense and debt variables, at the 2.5th and 97.5th percentiles to reduce the influence of extreme accounting realizations before computing the implied rate.

After constructing the implied interest rate, we restrict the sample to observations with values less than or equal to 30 percent. This upper bound eliminates implausibly high rates that are typically driven by very small denominators or transitory accounting irregularities rather than genuine borrowing costs. Following this truncation step, we winsorize the interest rate distribution at the 5th and 95th percentiles, consistent with standard practice in the corporate finance literature, including [Pittman and Fortin \(2003\)](#). This symmetric winsorization further limits the influence of residual outliers while preserving economically meaningful variation in borrowing costs.

The same trimming and winsorization strategy is applied to an alternative interest rate measure based on total liabilities, which serves as a robustness check in the empirical analysis. Together, these filtering steps ensure that the resulting cost of debt measure reflects economically plausible financing conditions and is not driven by mechanical accounting artifacts or

extreme observations.

C.2.2 Measuring Productivity

Firm level productivity is measured using a standard residual approach, following [Solow \(1957\)](#). Total factor productivity $Z_{i,t}$ for firm i at time t is computed as:

$$Z_{i,t} = \log(Y_{i,t}) - [\beta \log(H_{i,t}) + (1 - \beta) \log(K_{i,t})], \quad (104)$$

where $Y_{i,t}$ denotes firm level value added, $H_{i,t}$ represents labor input, $K_{i,t}$ denotes capital input, and β is the labor share of income. We use industry year specific labor and capital cost shares from the NBER CES database to construct this measure, following [Becker et al. \(2021\)](#). Under the assumptions of constant returns to scale, cost minimizing firms, and competitive factor markets, observed factor cost shares equal output elasticities, so that the Solow residual provides a valid measure of revenue total factor productivity.

Under these assumptions, productivity can be recovered directly from observed value added and factor inputs without estimating a production function, which avoids potential biases associated with simultaneity and functional form misspecification.